Observer-Patch Holography as a String-Vacuum Selector: Observers, Clocks, Edge Strings, and the Bouchard-Donagi Candidate

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Abstract

String theory is read here as an effective language for Observer-Patch Holography (OPH) edge dynamics. Finite observer patches compare shared records, and settled repair histories become worldsheet-like histories on the edge system.

The paper uses that reading as a vacuum-selection sieve. Complete reversible response and endogenous overlap transport force the $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ Lie type. Icosahedral incidence and target-blind port readback derive the signed finite response. A conditional matrix-current and matter realization supplies normalized hypercharge, three colors, a common $\mathbb{Z}_6$ kernel, and the maximal faithful Standard Model matter image. No source selection of that realization or of its physical global quotient is claimed. Laboratory attachment, three-family attachment, scalar multiplicity, and exclusion of extra light sectors require separate inputs.

A conventional string vacuum passes the sieve only after geometric, worldsheet-critical, threshold, and moduli-locking tests. The worked example is a heterotic Bouchard–Donagi construction with one massless Higgs pair. Its published data give the visible massless-cohomology spectrum, while the stabilized nonzero-mode spectrum and low-energy decoupling map are absent. The smooth one-Higgs space has 71 complex pre-completion moduli. The five-coordinate comparison map has rank zero on that space and supplies no physical slice or Jacobian. The example is therefore a structural comparison, not a selected vacuum.

What This Paper Contributes

String theory supplies a large language of edge dynamics, compactification, moduli, and Standard Model candidates. This paper reads that language from the OPH side. The OPH core supplies the target spacetime, records, clocks, and observer-facing boundary data. Its complete-response theorem forces the local Standard Model gauge Lie algebra. The conditional matrix-current and matter realization supplies the normalized hypercharge lattice, $N_c = 3$, common $\mathbb{Z}_6$ kernel, and maximal faithful matter image. Source selection of the matrix current, matter action, and global quotient is not claimed; each is a premise. The one-Higgs branch, three-family attachment, and no-extra-light-sector conditions are declared selector inputs rather than consequences of that finite theorem. The string description then becomes an effective edge language for histories constrained by those stated inputs.

The sieve carries the interaction package with the particle spectrum: the OPH Einstein branch supplies the gravity target, the compact-gauge branch supplies the strong/electroweak Lie algebra, and the conditional matter theorem fixes the visible charge lattice and maximal faithful image. The string representative must realize those forces and the particle content on the same observer-visible branch.

The worked contribution is a sieve for string-vacuum candidates. A Bouchard–Donagi type heterotic object is treated as a named structural test row and tested against six necessary conditions: one massless Higgs pair, a compatible Standard Model global form, operator safety, worldsheet criticality, threshold compatibility, and moduli locking. Every unproved condition is stated as a premise. The moduli calculation gives a negative rank and isolation result, rather than only a target equation. A vacuum survives only if its string data land on the OPH quantitative vector without retuning and all physical source directions are locked. The statement tests string data against these conditions; it does not turn a reference ensemble or ordinary string vacuum into the OPH-native vacuum.

Contents

1 Scope and Claim Boundary

This paper records a specific bridge from the de Sitter observer problem to the OPH string continuation. It separates three layers:

$$\begin{aligned}&\text{observer/clock implementation}\\ &\quad\longrightarrow \text{edge-string effective language}\\ &\quad\longrightarrow \text{critical string representative.}\end{aligned}$$

The first layer is OPH fixed-cutoff patch algebra. The second layer is the heat-kernel edge-sector read-off. The third layer is the Bouchard-Donagi heterotic Standard Model used as a named heterotic zero-mode candidate. The heat-kernel edge identity by itself does not prove a critical worldsheet CFT; the critical-edge certificate below is an additional condition on the continuum edge branch.

1.1 Forces, gauge group, and effective strings

The reader-facing phrase "particles and forces" separates into distinct mathematical conditions. The observer-visible normal form supplies representation and charge data; explicit action and phase hypotheses supply classical carrier modes; and a physical Hilbert space with a positive-residue pole supplies a particle. None of the latter two steps follows from an abstract reconstructed group.

Gravity. The OPH target is a single Einstein-frame metric on the physical quotient whose pure-Einstein, flat-background quadratic action has two transverse-traceless classical null modes. The critical representative must supply the quantum-particle receipt for a closed-string symmetric spin-two state, identify its classical limit with those metric modes, and match $G_4^{\text{string}} = G_4^{\text{OPH}}$.

Strong force. Conditional on the explicit response and rank-15 matter contracts, the color factor is the $\mathrm{SU}(3)_c$ part of the global quotient $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$. The visible compactification must realize this charge lattice with no light chiral exotics.

Weak and electromagnetic forces. The finite electroweak target, conditional on the explicit response and rank-15 matter contracts, is the $\mathrm{SU}(2) \times \mathrm{U}(1)$ branch and exact hypercharge lattice. The one-Higgs condition and the resulting low-energy $U(1)_{\text{em}}$ are separate premises. The candidate must land on that declared one-Higgs Standard Model branch and preserve the contract-conditional OPH hypercharge quotient. The local Borel–Weil model $H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2$ is used only for the observed light one-Higgs electroweak projection; it does not replace the full supersymmetric Higgs-pair bookkeeping in a string witness.

Standard Model local gauge algebra and conditional matter image. Complete reversible response and endogenous overlap transport force the local Standard Model gauge Lie algebra. Incidence and target-blind port readback derive the signed response. Under the conditional matrix-current and rank-15 matter contracts, the tensor calculation gives the maximal faithful matter image $G_{\text{packet}} = (\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$ inside the declared finite menu. The cover and its three nontrivial central quotients carry the same local tensors. The six-axis calculation matches the $\mathbb{Z}_6$ menu only after its coefficient relations are declared. It does not select the physical quotient. Laboratory current and flux identification and the continuum global form require separate physical attachments. The Bouchard-Donagi Wilson-line centralizer must equal $S(\mathrm{U}(3) \times \mathrm{U}(2)) \cong G_{\text{packet}}$, excluding simple-GUT X/Y generators from the unbroken visible adjoint. Connection dynamics and particles require separate receipts.

Effective string theory. The OPH input is stable cyclic edge normal forms, collar sewing, records, and accepted repair histories. The fixed-cutoff read-off is $Z_{\text{edge}} = K_t(1)$; a controlled large-edge branch gives the perturbative worldsheet language, and the critical-edge certificate decides whether the branch is a heterotic critical string.

Thus the paper does not add forces after selecting particles. It first fixes the observer-visible interaction structure and then asks which string compactification realizes that whole structure. The three Standard Model gauge forces are the low-energy decomposition of the global quotient, while the pure-Einstein tensor carrier is read as a closed-string graviton only after a critical lift supplies the required physical-state and pole receipts.

The selection statement is conditional. The rank certificate shows that its antecedent fails for the stated BD source and target data. Consequently the selector admits no string representative from this row.

The notation separates two roles:

$$BD^{\text{OPH}}_{n=1} \quad \text{labels the geometric one-Higgs structural row,}$$

$$BD^{\text{OPH}}_{n=1,+} = BD^{\text{OPH}}_{n=1} + \mathbb{Z}_4^R \quad \text{labels the tested operator-safe proposal.}$$

The $+$ records the MSSM $\mathbb{Z}_4^R$ safety layer. The charge-algebra statement is proved inside the paper; realization of $\mathbb{Z}_4^R$ as a symmetry of the BD compactification is a separate premise. The transverse-rank obstruction of Theorem 16.7 rejects this proposal as a selected witness. The underlying BD zero-mode construction is a structural benchmark.

The compact rule is:

$$\text{string theory is the edge-language of OPH.}$$

In that reading, the string landscape is the space of effective edge-worldsheet completions. OPH acts as a selector by imposing the observer-visible normal form.

The contrast with landscape reasoning is structural. Each declared pixel readout map has exactly one fixed point on its declared domain, as certified by the machine-checked domain-global uniqueness certificate in the public artifact bundle. This is uniqueness within a supplied map, not selection of a unique physical pair (P, N). Physical promotion of a pixel root requires same-scheme transport, target-independent map selection, and a typed same-quantity bridge. Capacity promotion has its own source-readback and attachment premises. On the OPH side of the map there is no vacuum landscape to relocate into, and the fixed target packet carries no retunable target-side dials. The BD source does not determine many physical continuous directions; the rank certificate below makes that distinction explicit. The sieve reading of string theory is therefore an edge-language statement, a statement about which worldsheet completions can present the declared target normal form rather than a competing selection mechanism, and it inherits no anthropic weighting from the space of candidates it filters.

Question	Statement	Evidence
Named string test row	OPH tests the operator-safe Bouchard–Donagi one-Higgs heterotic Standard Model proposal $BD^{\mathrm{OPH}}_{n=1,+}$. It fails selection under the moduli-rank obstruction.	Selector conditions, global group proof, safety charge table, structural sieve, and negative rank certificate.
String-theory pressure points	The landscape, global group, Higgs multiplicity, exotics, operator danger, proton decay, and moduli problems become explicit conditions.	Necessary-condition tables and falsification criteria.
Empirical test surface	The paper defines low-energy and threshold proxy targets against which a completed branch could fail.	Higgs/top target coordinates, stop proxy, one-loop gauge-running proxy, operator table.

2 OPH Entry Map for String Theorists

For a reader entering OPH from string theory, the shortest safe translation is this. OPH begins with finite observer patches. Each patch has local data, records, and overlap readouts. A physical world is the quotient-normal form reached when accepted repair moves lower mismatch and the local-diamond plus repair-completeness conditions make the observer-facing result schedule-independent from a fixed initial quotient state. Same-boundary uniqueness requires a preserved boundary or sector with a unique consistent quotient extension. Geometry, gauge structure, particles, and edge strings are read from that normal form.

2.1 Five informal summaries

1. The basic rule. Observer patches are finite. They see shared boundary data. Physics is the public content that survives overlap comparison and repair. The synthesis paper gives the broad entry point [1]; the consensus paper gives the fixed-cutoff repair theorem surface [3].

2. Why spacetime and gauge theory appear. Null structure, Lorentzian geometry, and the Einstein branch are recovered on the stated observer-overlap consistency surface. In the gauge branch, overlap/holonomy data classify fixed-stage zero-obstruction transportable sectors; on a cofinal tail carrying the compact-gauge refinement receipt, DR/Tannaka reconstruction gives a compact group from that category. Complete reversible response and endogenous overlap transport force the local Standard Model gauge Lie algebra. Incidence expresses the antipode J as a polynomial in adjacency, while target-blind impulse and port readback derive the signed response. Under the conditional matrix current and declared fermionic Spin category, the finite anomaly, determinant, spin-descent, and central-kernel calculations give the normalized hypercharge lattice, $N_c = 3$, and the maximal faithful matter image inside the declared finite menu. The local tensors also descend through the cover and its $\mathbb{Z}_2$, $\mathbb{Z}_3$, and $\mathbb{Z}_6$ quotients. The six-axis flux menu matches the $\mathbb{Z}_6$ quotient after its relations are declared. It does not select the physical global form. Laboratory current and flux identification, four-dimensional instanton normalization, and theta periodicity require separate physical premises. Here obstruction cancellation means both strictifying the central or higher associator defect and finding at least one allowed strict edge 1-cocycle representative with trivial represented holonomy. That combined criterion supplies transportability and classification. Three-family attachment, scalar multiplicity, and exclusion of extra light sectors are separate assumptions.

The stated premises select none of them. The Standard Model gauge paper carries the theorem surface for this claim [2]. The Yang-Mills note states the compact-gauge branch in the language of holonomy, Euclidean transfer, and the mass-gap problem [8].

3. Why particle numbers enter the selector. The particle-sector paper carries the electroweak, Higgs/top, quark, charged-lepton, neutrino, and hadron comparison surfaces [5]. The fine-structure note isolates exact roots of declared local pixel maps that enter the conditional quantitative electroweak target [6]. These papers supply the numerical vector against which the named BD structural row is screened.

4. Why observers and records are literal inputs. The screen-microphysics paper defines a candidate finite record-bearing carrier model with ports, interfaces, synchronization rules, and checkpoint readouts [4]. Its current two-carrier reference fixture has no composable seam triangle, so its higher-overlap Čech condition is vacuous and it does not certify a physical federation, source instrument, or carrier-to-support S^2 map. The thinking paper shows the same patch-net fixed-point machine as a model of biological cognition [9]. For this string paper, record-bearing finite observers are a stated input, not a result of the finite reference fixture.

5. Why phenomenology becomes a list of physical conditions. The dark-matter paper treats missing gravitational response as a quotient-edge scalar-channel phenomenon with explicit stress-parent conditions [7]. The metaphysics continuation layer is background for the fixed-point ontology and is outside the string selector proof.

2.2 Related OPH results

Ten companion results supply the inputs used here.

Source	Role in this paper	Citation
From Observer Consensus to Standard Physics	Synthesis entry point: finite observer cuts, overlap agreement, screen-capacity language, and the recovered-branch map.	[1]
Deriving Standard Model Gauge Structure from Observer Overlap Consistency	Full categorical and finite-carrier gauge routes, the axiom-forced local Standard Model gauge Lie algebra, and the conditional matrix-current hypercharge lattice, color count, common kernel, and maximal faithful matter image. Source selection of the matrix current, matter action, and physical global quotient is not claimed. Family attachment, scalar multiplicity, and extra-light-sector exclusion are separate assumptions; there is no mixed X/Y generator in the connected visible adjoint.	[2]

Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics	Fixed-cutoff consensus theorem: mismatch functional, accepted repair, quotient confluence, and holonomy obstructions.	[3]
Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH	Candidate finite carrier model for records, ports, interfaces, checkpoint restoration, and observer synchronization. The present two-carrier fixture has only vacuous higher-overlap closure and does not certify a physical federation, source instrument, or support-S^2 map.	[4]
Deriving the Particle Zoo from Observer Consistency	Particle and electroweak quantitative branches used by the moduli-locking target vector.	[5]
The Fine-Structure Constant as an OPH Pixel Fixed Point	Local screen-cell fixed point and fine-structure input used by the electroweak target.	[6]
Observer-Patch Holography and the Dark Matter Phenomenon	Quotient-edge scalar-channel branch, used as a phenomenology example of OPH conditional logic.	[7]
Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics	Receipt-certified compact-gauge reconstruction and mass-gap route, useful for the gauge side of the string selector.	[8]
Thinking as Patch-Net Fixed-Point Search	Cognitive instance of the same finite patch-net record and repair machine.	[9]

3 Main Result: Conditional BD Witness Selection up to OPH Equivalence

An OPH-correct critical string is defined only after the equivalence relation has been fixed. The claim concerns observer-visible physical data, not notation, duality frame, worldsheet gauge, or coordinate presentation. This paper supplies a named BD witness target and a finite encoded comparison set. A global singleton statement requires an exhaustive candidate-class certificate in addition to the witness certificates.

Definition 3.1 (OPH-equivalent critical presentations)**.** *Two critical-string presentations $\mathcal{T}_1$ and $\mathcal{T}_2$ are OPH-equivalent, written*

$$\mathcal{T}_1 \sim_{\mathrm{OPH}} \mathcal{T}_2,$$

when they induce the same observer-visible data:

1. *the same edge-sewn heat-kernel partition on the compact gauge branch;*
2. *the same four-dimensional tensor-mode normalization and Einstein-frame metric perturbation;*
3. *the same global visible gauge group*

$$G_{\mathrm{phys}} = (\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6;$$

4. the same hypercharge lattice, chiral index, and low-energy matter package;

5. the same operator-safety algebra on the visible superpotential;

6. the same complete OPH quantitative readout, including physical threshold and modulus effects.

Here "OPH-invisible" means an independently established presentation redundancy: a diffeomorphism or bundle/B-field gauge orbit, bundle isomorphism, local representative or exact field-basis change, or a verified scheme/duality change with identical complete public readout. It is not defined as the kernel of a chosen finite target map. Physical scalar directions cannot be discarded merely because the fixed five-coordinate packet does not read them. Duality frames are charts on the same OPH-visible normal form only after that equivalence has been proved.

Lemma 3.2 (Quotient descent and presentation invariance)**.** *Let a redundancy group G preserve the completed constraint locus, and let every component of the observer-visible readout be G-invariant. The constraints and readout then descend to each fixed-orbit-type stratum of the physical quotient. If two quotient charts are related by a C^1 diffeomorphism ϕ, and two equivalent output conventions are related by a C^1 diffeomorphism ψ, then*

$$F' = \psi \circ F \circ \phi^{-1}, \qquad DF' = D\psi\, DF\, D\phi^{-1}.$$

Thus rank on the physical tangent, kernel dimension, and local isolation are independent of the certified chart or convention. A scheme change or string duality can be placed in the invisible quotient only after the transition maps and equality of the complete physical readout are proved. An unclassified continuous stabilizer, singular orbit-type change, or gauge-copy ambiguity fails this lemma's hypotheses and must be treated on a separate stratum.

Proof. Invariance makes the constraints and readout constant on every G-orbit, so the universal property of the quotient gives the descended maps. The derivative formula is the chain rule. Both outer derivatives are invertible, which preserves rank and kernel dimension. Diffeomorphisms also preserve whether a target fiber is locally a singleton. The final sentence records cases in which no such quotient chart has been certified. □

Definition 3.3 (OPH-correct critical string)**.** *A critical-string presentation $\mathcal{T}$ is OPH-correct if it satisfies all of the following.*

1. a critical completion of the OPH sewn-edge worldsheet branch, with the same observer-visible edge data.

2. Its BRST-physical massless spin-two state supplies a positive-residue quantum completion of the OPH Einstein-frame classical tensor mode.

3. Its visible compact gauge sector descends to $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$.

4. Its visible chiral spectrum is exactly three generations, no light chiral exotics, and the one-Higgs-pair branch used by the OPH electroweak target.

5. Its visible operator algebra has the $\mathbb{Z}_4^R$ safety layer, or an OPH-equivalent rule, which permits Yukawas and the Weinberg operator, and forbids perturbative RPV, perturbative dimension-five proton decay, and the perturbative μ-term.

6. *Its completed physical equations have a dynamically stable solution, and its precommitted scheme-locked threshold map satisfies*

$$\mathcal{F}_{\mathcal{T}}(m_\star) = \mathcal{O}_{\mathrm{OPH}}$$

with full transverse rank jointly with the completion constraints after quotienting independently proved OPH-invisible redundancies.

Theorem 3.4 (Conditional BD witness theorem)**.** *Assume the contract-conditional OPH finite gauge target, together with declared selector inputs $N_g = 3$, one light Higgs doublet, no light chiral exotics, and no extra visible low-scale* U(1)*. Assume the heterotic edge-sector branch including the finite-carrier critical-edge certificate and the BD conditions listed in Section 19. Assume also that the Bouchard–Donagi one-Higgs branch with the operator-safety layer satisfies the cohomology, global-group, safety, threshold, and moduli-locking conditions, including a certified low-energy decoupling map. Then*

$$\boxed{BD^{\mathrm{OPH}}_{n=1,+}}$$

is a passing named OPH-correct critical-string witness, modulo $\sim_{\mathrm{OPH}}$.

Proof. The complete-response theorem forces the local Standard Model gauge Lie algebra. Under the conditional matrix current and declared fermionic Spin category, the matter theorem fixes three colors, the exact hypercharge lattice, the maximal faithful matter image, and the product-adjoint absence of mixed X/Y connected generators inside the declared finite menu. Source selection of the matrix current, matter action, and physical $\mathbb{Z}_6$ quotient is not claimed. The theorem does not fix laboratory attachment, family attachment, scalar multiplicity, or extra-light-sector exclusion. Those conditions enter this theorem as declared selector inputs. The edge-string continuation restricts the witness to critical presentations whose worldsheet branch is the sewn OPH edge system and whose BRST-physical massless spin-two state has the OPH metric perturbation as its classical limit. The Bouchard–Donagi $\mathbb{Z}_2$-quotient SU(5)-bundle with one Higgs pair supplies the named heterotic witness with the required three-generation, no-exotics visible massless cohomology and Wilson-line centralizer. The $\mathbb{Z}_4^R$ layer supplies the visible operator condition. The assumed threshold/spectrum and decoupling certificate supplies the physical low-energy map. The moduli-locking hypothesis isolates the physical target point for that witness. Those conditions define OPH-correctness for the named witness. Quotienting by $\sim_{\mathrm{OPH}}$ removes pure presentation duplicates of the same observer-visible data. □

Corollary 3.5 (BD selection scope)**.** *The antecedent of the conditional witness theorem is false for the published input data. In particular, no completed constraint system, dynamically stable point, physical readout, or constraint-augmented Jacobian is supplied. The locking and isolation condition is obstructed by Theorem 16.7. Hence*

$$BD^{\mathrm{OPH}}_{n=1,+} \quad \textit{is not a selected or passing OPH-string witness.}$$

The published BD zero-mode construction is a structural benchmark. The independent OPH recovered-core theorems have different premises.

Proposition 3.6 (Catalogue-relative singleton certificate criterion)**.** *Let $\mathcal{U}$ be a declared candidate universe and let $\mathcal{C}$ be an explicitly enumerated catalogue. Assume a coverage certificate proves that every element of $\mathcal{U}$ is represented by a row of $\mathcal{C}$ up to $\sim_{\mathrm{OPH}}$. Assume also that every discrete branch and every physical parameter domain has a certified cover with no unresolved cells, and that the* *candidate conditions are recomputed for every resulting equivalence class. If the exact passing set is the single class $BD^{\mathrm{OPH}}_{n=1,+}$, then*

$$\boxed{\{\mathcal{T} \in \mathcal{U} : \mathcal{T} \text{ is OPH-correct}\}/\sim_{\mathrm{OPH}} = \{BD^{\mathrm{OPH}}_{n=1,+}\}.}$$

Without the coverage certificate, the same calculation makes a claim only about the explicitly enumerated catalogue. A minimality score does not remove a second physically passing class unless that ordering is independently derived as part of the OPH target.

Proof. Coverage maps every candidate in $\mathcal{U}$ to a certified catalogue row. The branch and parameter covers classify every row as passing or failing, with no unclassified region. Quotienting the exact passing set by the certified equivalence relation gives the stated singleton by hypothesis. If catalogue coverage is absent, the same argument has no premise for candidates outside $\mathcal{C}$. A score cannot change membership in the passing set unless its ordering is itself one of the independently fixed physical conditions. □

Remark 3.7 (How strong this is). BD is not decoration here. OPH supplies an external observer-visible target, so a named branch can be tested by a normal-form equation. The claim is falsifiable by construction: failure of the BD cohomology, safety-layer realization, threshold/spectrum certificate, decoupling map, or transverse moduli rank excludes the named string proposal while leaving the OPH recovered core intact. The moduli certificate realizes that failure case. Global uniqueness is a separate comparative-catalogue question, moot for this row because it does not satisfy the existence conditions.

4 Conditional Theorem Structure

The de Sitter observer literature isolates a compact set of mathematical questions. The corresponding OPH statements are:

Pressure point	**OPH statement**
Physical observer in de Sitter	The finite-observer premise identifies a physical observer with a stable record-bearing patch subfederation.
Complex semiclassical correlators	The clock-projection correlator statement places imaginary phases after record-sector clock projection.
Time-reversal holonomy	Clock flips are $\mathbb{Z}_2^T$ cycle obstructions, equivalently $w_1(L_T)$ classes.
Time-reversal breaking	Ordered checkpoint records select a local time orientation.
Entropy location	The edge-capacity identification reads $S_{\mathrm{dS}} = A/(4\ell_P^2)$ as finite edge-center record capacity.
Finite de Sitter degrees	Finite patch and interface algebras supply pre-geometric matrix degrees.
DSSYK/QCD/string parallels	Heat-kernel edge sums reorganize as large-N worldsheet expansions.
Critical worldsheet closure	A heterotic interpretation requires the sewn-edge scaling limit to realize $(\mathrm{Heis}_8 \otimes V_{\Gamma_{16}})_L \otimes (\mathrm{Heis}_8 \otimes \mathrm{Ferm}_8)_R$, with no residual coset.

The string and vacuum statements use the following conditions:

Pressure point	OPH statement
String landscape	The vacuum sieve requires critical completions to hit the observer-visible normal form.
Worldsheet criticality	A critical interpretation requires the sewn-edge scaling limit to be identified with the heterotic edge VOA, including OPEs, grading, spin structures, and no residual coset.
Global Standard Model group	The global group identity is $S(\mathrm{U}(3) \times \mathrm{U}(2)) \cong (\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$.
Connected unified-gauge adjoint	The product-group adjoint contains no mixed $(3, 2, \pm 5/6)$ generator; connection dynamics and particle spectra require separate premises.
Moduli stabilization	The moduli-locking criterion is $\mathcal{F}(m_\star) = \mathcal{O}_{\mathrm{OPH}}$ with full transverse rank.

5 De Sitter Observer Pressure Points

Two de Sitter papers make the observer problem unusually explicit. One note relates the need for observers in de Sitter space to spontaneous breaking of time reversal [10]. A follow-up states that quantum de Sitter theory is widely thought to require a physical observer in the static patch, with the definition of observer unspecified [11]. The time-reversal question is sharpened by treating time reversal as a gauge symmetry hidden by spontaneous symmetry breaking, with a proposed "smoking gun" given by a closed curve whose holonomy flips forward-going clocks into backward clocks [12].

OPH reads these as implementation questions. What finite object counts as an observer? What operation selects the clock branch? What finite obstruction carries the clock flip? What entropy surface supplies the static-patch degrees of freedom?

5.1 Observer

Definition 5.1 (OPH observer subfederation)**.** *Let*

$$\mathfrak{F} = (V, E, \{\mathcal{A}_i\}, \{\mathcal{I}_e\}, \{\pi_{i,e}\}, \{\mathcal{R}_i\}, \{\mathcal{U}_i\})$$

be a finite OPH patch carrier. An observer subfederation is a finite $U \subset V$ equipped with an accessible algebra $\mathcal{A}_U$, a state ρ_U, a record algebra $\mathcal{R}_U$, a boundary interface $\mathcal{I}_{\partial U}$, and accepted repair maps whose normal form preserves a checkpoint order on $\mathcal{R}_U$.

The observer projection is the record event

$$P_{\mathrm{obs}} = P_{\mathcal{R}_U,+},$$

where $+$ denotes the chosen clock orientation. Conditioning is ordinary projection:

$$\rho|_{\mathcal{R}_U,+} = \frac{P_{\mathcal{R}_U,+} \rho P_{\mathcal{R}_U,+}}{\mathrm{Tr}(P_{\mathcal{R}_U,+}\rho)}.$$

Theorem 5.2 (Finite observer theorem)**.** *On a finite OPH static-patch carrier, a physical observer is a subfederation $U \subset V$ whose checkpoint*

$$\mathrm{Chk}_U(t) = \left(\mathcal{R}_U(t), \rho_U^{\mathrm{acc}}(t), \mathfrak{I}_U^{\mathrm{ext}}(t), \nu_{\geq t}, \mathfrak{B}_U(t)\right)$$

has a stable continuation law on the observer-accessible event algebra under same-interface continuation. The de Sitter observer projection is the record-sector projection $P_{\mathcal{R}_U,+}$.

Proof. The data defining U are finite algebraic data. The event that records persist with a monotone checkpoint ordering is a projection in the record algebra after passing to the observable quotient. Conditioning by that projection gives the usual post-selected state on the observer-accessible sector. The observer is represented inside the patch Hilbert space. □

5.2 Semiclassical De Sitter

OPH treats semiclassical spacetime as an observer-facing quotient normal form. The repair functional has the schematic form

$$\Phi(s) = \sum_{e=\{i,j\}} w_e \, d_e(\pi_{i,e}(s_i), \pi_{j,e}(s_j)) \,.$$

Accepted repairs lower Φ. Under the local-diamond and repair-completeness hypotheses used in the consensus paper, the terminal observable state is schedule-independent on the physical quotient.

Claim 5.3 (Semiclassical patch)**.** *The semiclassical de Sitter patch is the stable quotient normal form of finite observer-overlap repair, read on a record-bearing clock sector.*

OPH answers the de Sitter observer question here. The maximally mixed static-patch state belongs to the unconditioned algebra. The semiclassical branch is read after record conditioning.

5.3 Clock-conditioned correlator

The finite toy model used in the correspondence package has two clock orientations exchanged by time reversal. The two-level model is only a witness. The algebraic point is that the maximally mixed static-patch trace averages over clock orientations, and an observer record projection selects one oriented branch.

Theorem 5.4 (Clock-projection correlator theorem)**.** *Let $\mathcal{H}$ be finite-dimensional, let $H = H^\dagger$, let $A = A^\dagger$, and set*

$$A(t) = e^{iHt} A e^{-iHt}, \qquad \rho_\infty = I/d.$$

Then

$$C_\infty(t) := \mathrm{Tr}(\rho_\infty A(t) A)$$

is real, so

$$\mathrm{Im}\, C_\infty(t) = 0.$$

Equivalently,

$$\mathrm{Tr}(\rho_\infty [A(t), A]) = 0.$$

For a noncentral observer-clock record projector P_E,

$$\rho_E = \frac{P_E \rho_\infty P_E}{\mathrm{Tr}(P_E \rho_\infty)},$$

the commutator expectation is generically nonzero. In the two-state clock example

$$H = \frac{\omega}{2} \sigma_z, \qquad A = \sigma_x,$$

one has

$$C_\infty(t) = \frac{1}{2} \mathrm{Tr}(\sigma_x(t) \sigma_x) = \cos(\omega t).$$

On the forward clock record state $\rho_+ = |0\rangle\langle 0|$,

$$C_+(t) = \langle 0|\sigma_x(t)\sigma_x|0\rangle = e^{i\omega t}.$$

The imaginary semiclassical phase is restored by clock-record conditioning:

$$\operatorname{Im} C_\infty(t) = 0, \qquad \operatorname{Im} C_+(t) = \sin(\omega t).$$

Proof. Since $A(t)$ and A are Hermitian,

$$C_\infty(t)^* = \frac{1}{d}\operatorname{Tr}(AA(t)).$$

By cyclicity of the trace,

$$\operatorname{Tr}(AA(t)) = \operatorname{Tr}(A(t)A),$$

so $C_\infty(t)^* = C_\infty(t)$, hence $C_\infty(t) \in \mathbb{R}$. The commutator statement follows from $\operatorname{Tr}([A(t), A]) = 0$.

For a projected state ρ_E, the projector sits between the state and the clock-transition operator. The cyclic cancellation above applies only in the commuting case. Generic observer-clock records are noncentral, so the antisymmetric part can survive. For the two-state witness,

$$\sigma_x(t) = e^{i\omega t\sigma_z/2}\sigma_x e^{-i\omega t\sigma_z/2} = \cos(\omega t)\sigma_x - \sin(\omega t)\sigma_y.$$

Then

$$\sigma_x(t)\sigma_x = \cos(\omega t)I + i\sin(\omega t)\sigma_z.$$

The maximally mixed trace gives $\cos(\omega t)$. The $|0\rangle$ matrix element gives $\cos(\omega t) + i\sin(\omega t) = e^{i\omega t}$. □

5.4 Time-reversal holonomy

The time-orientation data must be a local system. Globally chosen patch signs give only a coboundary, and every closed-cycle product then telescopes to $+1$. Clock-flip holonomy needs a $\mathbb{Z}_2^T$-valued Čech 1-cocycle on the overlap nerve $N(\mathfrak{F})$:

$$g_{ij} \in \mathbb{Z}_2^T, \qquad g_{ij}g_{jk}g_{ki} = 1$$

on triple overlaps, modulo local redefinitions

$$g_{ij} \sim \lambda_i g_{ij}\lambda_j^{-1}, \qquad \lambda_i \in \mathbb{Z}_2^T.$$

For a closed path $\gamma = (i_0 i_1)(i_1 i_2)\cdots(i_{n-1}i_0)$, define

$$h_T(\gamma) = \prod_{a=0}^{n-1} g_{i_a i_{a+1}} \in \mathbb{Z}_2^T.$$

Theorem 5.5 (Time-reversal holonomy theorem)**.** *A forward-going local clock transported around γ returns backward-going iff*

$$h_T(\gamma) = -1.$$

The obstruction to a global time-orientation section is the cohomology class

$$[g] \in H^1(N(\mathfrak{F}), \mathbb{Z}_2) \cong \check{H}^1(N(\mathfrak{F}), \mathbb{Z}_2),$$

equivalently the first Stiefel-Whitney class

$$w_1(L_T) = [g].$$

It vanishes iff the transition cocycle is a coboundary, i.e. iff one can choose local orientations ϵ_i with $g_{ij} = \epsilon_i^{-1}\epsilon_j$ on every overlap.

Proof. Parallel transport across the overlap (ij) multiplies the clock-orientation basis by g_{ij}. Transport around γ multiplies by the ordered product $h_T(\gamma)$. Since the group is $\mathbb{Z}_2$, the value $+1$ preserves orientation and the value -1 reverses it.

A global time orientation is a choice of signs $\epsilon_i \in \mathbb{Z}_2$ such that the transition from patch i to patch j is reproduced by $g_{ij} = \epsilon_i^{-1}\epsilon_j$. This is the statement that the 1-cocycle g is a coboundary. The obstruction class is $[g] \in H^1(N(\mathfrak{F}), \mathbb{Z}_2)$, which is the first Stiefel-Whitney class of the associated $\mathbb{Z}_2$ time-orientation line bundle L_T. A nonzero class means no global time-orientation section exists. □

The proposed de Sitter clock-flip test reads, in finite OPH form, as a $\mathbb{Z}_2^T$ cycle obstruction on the patch-overlap nerve.

5.5 Record-branch time-reversal breaking

Let $P_1, P_2, \ldots, P_n \in \mathcal{R}_U$ be mutually commuting checkpoint record projectors over a finite observation window, with a monotone checkpoint order

$$P_1 \prec P_2 \prec \cdots \prec P_n.$$

Time reversal sends this ordered history to the reversed history. Conditioning on the finite branch projector

$$P_{\text{branch}} = P_1 P_2 \cdots P_n$$

selects one local time orientation on the observer-accessible quotient.

Theorem 5.6 (Record-branch time-reversal breaking)**.** *The unconditioned finite carrier carries the time-reversal-related pair of record orders. A stable checkpoint branch selects one order by*

$$\rho \mapsto \frac{P_{\text{branch}}\rho P_{\text{branch}}}{\operatorname{Tr}(P_{\text{branch}}\rho)}.$$

Time reversal is a redundancy of the unconditioned carrier and is hidden on the conditioned observer quotient by finite record ordering.

Proof. Because the checkpoint records belong to the observer-accessible record algebra, they are central or mutually commuting on the exact fixed-cutoff event surface used here. The finite product $P_{\text{branch}} = P_1 \cdots P_n$ is again a projector onto the event that the ordered record history occurs. The time-reversed branch is represented by the same unordered product together with the opposite continuation schedule. The unconditioned carrier contains both orientation-related continuations. Conditioning by P_{branch} and the continuation schedule selects one law on the observer-accessible quotient. On that quotient the reversed branch is outside the conditioned continuation law, so the time-reversal redundancy is hidden by record ordering. □

6 Entropy, Matrix Degrees, and Scale Separation

DSSYK/JT-de Sitter work places the stretched-horizon entropy at order Planck distance from the mathematical horizon [13]. OPH agrees with that direction. The entropy budget is finite edge-center and record capacity:

$$S_{\text{dS}} = N_{\text{scr}} = \frac{A_{\text{dS}}}{4\ell_P^2}.$$

The string scale is downstream in this paper. The edge record capacity is the substrate; the worldsheet is the effective language after sewing and large-edge organization.

Theorem 6.1 (Edge-capacity entropy theorem)**.** *On the OPH de Sitter static-patch branch, the horizon entropy is the capacity of the observer-facing edge-center record surface:*

$$S_{\mathrm{dS}} = N_{\mathrm{scr}} = \frac{A_{\mathrm{dS}}}{4\ell_P^2}.$$

The corresponding screen-capacity branch gives

$$\Lambda = \frac{3\pi}{G N_{\mathrm{scr}}}.$$

Proof. The OPH screen-capacity branch identifies the number of observer-facing horizon record units with the Gibbons-Hawking entropy,

$$N_{\mathrm{scr}} = \frac{A_{\mathrm{dS}}}{4\ell_P^2}.$$

For a four-dimensional de Sitter static patch,

$$A_{\mathrm{dS}} = 4\pi R_{\mathrm{dS}}^2, \qquad \Lambda = \frac{3}{R_{\mathrm{dS}}^2}.$$

This gives

$$N_{\mathrm{scr}} = \frac{4\pi R_{\mathrm{dS}}^2}{4\ell_P^2} = \frac{\pi R_{\mathrm{dS}}^2}{\ell_P^2}.$$

Using $\ell_P^2 = G$ in units $\hbar = c = 1$,

$$N_{\mathrm{scr}} = \frac{3\pi}{G\Lambda},$$

and

$$\Lambda = \frac{3\pi}{G N_{\mathrm{scr}}}.$$

□

The finite carrier

$$\mathfrak{F} = (V, E, \{\mathcal{A}_i\}, \{\mathcal{I}_e\}, \{\pi_{i,e}\}, \{\mathcal{R}_i\}, \{\mathcal{U}_i\})$$

also supplies a matrix-style pre-geometric object. This matches the pressure behind de Sitter-matrix arguments, where black holes and nonperturbative sectors probe the static-patch degrees of freedom [14]. In OPH, black-hole sectors are constrained normal forms that reallocate screen capacity:

$$P(\text{sector}) \propto \exp[-(S_{\mathrm{dS}} - S_{\mathrm{sector}})].$$

Theorem 6.2 (Static-patch matrix carrier theorem)**.** *The finite algebras $\mathcal{A}_i$ and interface algebras $\mathcal{I}_e$ of an OPH de Sitter static-patch carrier are the microscopic matrix degrees used by the pre-geometric description. A black-hole sector is a constrained normal-form sector $\mathfrak{S}_{\mathrm{BH}}$ with*

$$S(\mathfrak{S}_{\mathrm{BH}}) < S_{\mathrm{dS}}$$

and fluctuation weight proportional to

$$\exp[-(S_{\mathrm{dS}} - S(\mathfrak{S}_{\mathrm{BH}}))].$$

Proof. At fixed cutoff each local patch algebra and interface algebra is finite-dimensional, hence a finite direct sum of matrix algebras. The unreduced tensor product of a finite static-patch carrier is a finite matrix algebra up to superselection-block decomposition, and the physical algebra is its overlap-invariant quotient. A constrained black-hole sector imposes additional horizon, energy, or area constraints on the same finite record surface, so its accessible record count is smaller than the empty de Sitter static-patch count. With the same coarse-grained entropy measure in both macroscopic sectors, the relative weight of the constrained sector is the ratio of state counts,

$$\frac{e^{S(\mathfrak{S}_{\mathrm{BH}})}}{e^{S_{\mathrm{dS}}}} = \exp[-(S_{\mathrm{dS}} - S(\mathfrak{S}_{\mathrm{BH}}))].$$

□

Scale-separation work on de Sitter and double-scaled SYK distinguishes a cosmic sector from a microscopic sector [15]. OPH has the same split:

$$\text{cosmic sector} \leftrightarrow N_{\mathrm{scr}}, \qquad \text{microscopic sector} \leftrightarrow P,\ \text{edge labels, finite records.}$$

The clean dimensionless separator is N_{scr}/P.

7 Edge-String Emergence

The OPH string continuation begins at the edge-sector partition. At fixed cutoff, edge labels are compact representations R with dimension d_R and quadratic Casimir $C_2(R)$. The open-edge weight is

$$p_R(t) = \frac{d_R e^{-tC_2(R)}}{Z_{\mathrm{open}}(t)}.$$

Sewing two collar sides contributes a second representation-dimension factor:

$$Z_{\mathrm{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}.$$

By Peter-Weyl,

$$K_t(g) = \sum_R d_R \chi_R(g) e^{-tC_2(R)}.$$

At $g = 1$, $\chi_R(1) = d_R$, hence

$$Z_{\mathrm{edge}}(t) = K_t(1).$$

Theorem 7.1 (OPH edge-string read-off)**.** *The sewn OPH collar partition equals the compact-group heat kernel at the identity:*

$$Z_{\mathrm{edge}}(t) = K_t(1).$$

On a large-N_{edge} branch with $g_s = 1/N_{\mathrm{edge}}$, the controlled surface expansion has the closed-string genus form.

Proof. The compact heat kernel has Peter-Weyl expansion

$$K_t(g) = \sum_R d_R \chi_R(g) e^{-tC_2(R)}.$$

At the identity, $\chi_R(1) = d_R$, so

$$K_t(1) = \sum_R d_R^2 e^{-tC_2(R)}.$$

A closed OPH collar is obtained by sewing two open edge boundaries, and the second boundary contributes the second dimension factor. The sewn collar partition is

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)} = K_t(1).$$

If a distinct large-edge branch has the controlled expansion

$$\log Z = \sum_g N_{\text{edge}}^{2-2g} F_g$$

with uniform remainder bounds, then $g_s = N_{\text{edge}}^{-1}$ rewrites the coefficient as

$$N_{\text{edge}}^{2-2g} = g_s^{2g-2},$$

the standard closed-string genus weight. □

Theorem 7.2 (Edge-sum universality theorem)**.** *On the controlled large-N_{edge} branch,*

$$\log Z_{\text{edge}} = \sum_g N_{\text{edge}}^{2-2g} F_g, \qquad g_s = \frac{1}{N_{\text{edge}}}.$$

The same representation sum gives the two-dimensional Yang-Mills heat-kernel basis and the world-sheet string basis.

Proof. The representation sum is the heat-kernel form of two-dimensional Yang-Mills. Its sewing law is the Chapman-Kolmogorov law for heat kernels, which is also the OPH collar-sewing law. The large-N_{edge} expansion reorganizes the same sewn surfaces by Euler characteristic:

$$N_{\text{edge}}^{2-2g} = g_s^{2g-2}.$$

The same edge representation sum has a 2D Yang-Mills heat-kernel reading and a perturbative closed-string worldsheet reading, provided the large-edge remainder bounds hold. □

This theorem is the bridge to the string community. It also explains why DSSYK, large-N QCD, and open-string structures keep meeting in the de Sitter flat-space limit. A 2025 DSSYK/QCD paper states that double-scaled SYK at infinite temperature and large-N QCD have large-N expansions of the same form, and that DSSYK provides a tractable window into the fixed-coupling flat-space limit [16]. A companion DSSYK flat-space paper connects the limit to strongly coupled $(1+1)$-dimensional QCD and open-string Regge behavior [17]. OPH reads this family of coincidences as different bases for sewn edge-sector sums.

8 What an OPH String Is

The heat-kernel identity gives the partition-function trace of a string. The object-level dictionary is finite. In OPH, a string is a stable one-dimensional edge-cycle normal form in the overlap carrier. A continuum string is the scaling description of that cycle after collar sewing and large-edge coarse graining.

Definition 8.1 (Fixed-cutoff OPH pre-string)**.** *Let*

$$\mathfrak{F} = (V, E, \{\mathcal{A}_i\}, \{\mathcal{I}_e\}, \{\pi_{i,e}\}, \{\mathcal{R}_i\}, \{\mathcal{U}_i\})$$

be a fixed OPH patch carrier. A fixed-cutoff pre-string is a cyclic collar word

$$\Gamma = (e_1, e_2, \ldots, e_n; e_{n+1} = e_1)$$

in the overlap nerve, together with edge labels and vertex intertwiners

$$\mathcal{S}_0(\Gamma) = (\Gamma, \{R_{e_a}\}_{a=1}^n, \{I_{v_a}\}_{a=1}^n, \{r_{e_a}\}_{a=1}^n),$$

where each R_e is a compact-gauge representation label carried by the edge-center algebra, each

$$I_v \in \mathrm{Hom}_G(R_{e_{a-1}} \otimes R_{e_a}, \mathbf{1} \oplus \cdots)$$

is the local fusion/sewing datum at a patch junction, and r_e is the record state exposed on the collar.

Definition 8.2 (OPH string)**.** *An OPH string is the quotient-normal-form class*

$$\mathcal{S} = [\mathcal{S}_0(\Gamma)]_{\mathrm{nf}} / (\textit{local repair, collar refinement, OPH-invisible relabeling}).$$

A closed string is a cyclic class with no external endpoint. An open string is an interval-class whose endpoints lie on declared observer branes, defect collars, or boundary record sectors. A multi-string state is a finite disjoint union of such classes, modulo the same repair quotient.

The cycle is visible through representation data and records on overlaps. Its support is one-dimensional in the overlap nerve. Its physical role comes from survival under local repair and refinement.

8.1 One string, many strings, and worldsheets

A single OPH string at one checkpoint is

$$\mathcal{S}(\tau_0) = [\Gamma(\tau_0), R(\tau_0), I(\tau_0), r(\tau_0)]_{\mathrm{nf}}.$$

A history of the same object through accepted repairs is a sequence

$$\mathcal{S}(\tau_0) \to \mathcal{S}(\tau_1) \to \cdots \to \mathcal{S}(\tau_m).$$

The two-dimensional worldsheet is the swept collar complex

$$\Sigma_{\mathcal{S}} := \bigcup_{j=0}^{m-1} \Gamma(\tau_j) \times [\tau_j, \tau_{j+1}],$$

with sewing at repair events. Pair-of-pants splitting and joining are OPH cobordisms in which one cyclic normal form changes into two, or two change into one, preserving exposed overlap constraints.

The dictionary is:

$$\begin{array}{ccl} \text{OPH cyclic edge normal form} & \longleftrightarrow & \text{string at one time,} \\ \text{accepted repair history of the cycle} & \longleftrightarrow & \text{worldsheet,} \\ \text{cycle split/merge cobordism} & \longleftrightarrow & \text{string interaction,} \\ \text{finite edge capacity} & \longleftrightarrow & g_s^{-1} \text{ on the controlled large-edge branch.} \end{array}$$

8.2 Why the string vibrates

The string vibrates because the cyclic edge normal form has internal normal modes. Linearize the repair law around a stable cyclic solution $\mathcal{S}_\star$. If ξ_a denotes a small edge-position or edge-label perturbation at the a-th collar site, the quadratic mismatch is

$$\Phi(\mathcal{S}_\star + \xi) = \Phi(\mathcal{S}_\star) + \frac{1}{2}\sum_{a,b} \xi_a K_{ab} \xi_b + O(\xi^3).$$

On a long uniform cycle, the Hessian K is a graph Laplacian plus local mass/curvature terms. The normal modes are Fourier modes

$$\xi_a^\mu(\tau) = \sum_{n\in\mathbb{Z}} \xi_n^\mu(\tau) e^{2\pi i n a/N}.$$

In the continuum edge limit, with $a/N \to \sigma/(2\pi)$, this becomes the closed-string field

$$X^\mu(\sigma,\tau) = x_0^\mu + p^\mu \tau + i\sqrt{\frac{\alpha'_{\mathrm{OPH}}}{2}} \sum_{n\neq 0} \frac{1}{n}\left(\alpha_n^\mu e^{-in(\tau-\sigma)} + \tilde{\alpha}_n^\mu e^{-in(\tau+\sigma)}\right).$$

The left/right split is the two-sided collar split. One side of the sewn edge supplies the left movers and the other supplies the right movers. Quantization of the finite symplectic record/edge normal modes gives the oscillator algebra in the scaling limit,

$$[\alpha_m^\mu, \alpha_n^\nu] = m\delta_{m+n,0}\eta^{\mu\nu}, \qquad [\tilde{\alpha}_m^\mu, \tilde{\alpha}_n^\nu] = m\delta_{m+n,0}\eta^{\mu\nu}.$$

At this stage the range of μ is not fixed by the heat-kernel theorem or by the oscillator form alone. The critical dimension and the heterotic internal central charge require the critical-edge certificate in Section 9. The parameter α'_{OPH} is the continuum normalization of the edge diffusion/repair time. Its identification with a physical string tension is conditional on matching the four-dimensional Newton constant, the OPH pixel scale, and a certified compactification threshold.

Theorem 8.3 (String emergence theorem)**.** *Assume a fixed-cutoff compact edge branch, collar sewing, a separated refinement system, and a controlled large-N_{edge} limit in which cyclic normal forms have a continuum embedding $X : \Sigma \to M_{\mathrm{OPH}}$. Then the OPH edge-cycle normal-form category maps to a perturbative string worldsheet category. The map sends cyclic edge normal forms to string states, accepted repair histories to worldsheets, and edge-cycle split/merge cobordisms to string interactions.*

Proof. At fixed cutoff, the object is finite: a cyclic word of edge collars with compact representation labels and intertwiners. Sewing gives the heat-kernel partition described above. Refinement turns long cyclic words into one-dimensional continua. A time-ordered repair history sweeps a cellulated two-complex. The large-edge expansion supplies the genus weighting. The normal-mode expansion of the repaired cycle gives the oscillator degrees of freedom. These are precisely the data used by a perturbative worldsheet description. □

Remark 8.4 (Claim boundary). The exact fixed-cutoff theorem is the edge heat-kernel/sewing theorem. The continuum string, oscillator algebra, and genus expansion require the large-edge/refinement hypotheses. OPH begins with finite edge data; string theory is the controlled language of its sewn-edge scaling branch.

9 Critical Central-Charge Condition

The declared integer-counting realization in the OPH screen and particle papers supplies the relevant integer targets for a heterotic branch [4, 5]. Those integers become central charges only under the conditions below. The screen architecture gives twelve primitive curvature/record ports,

$$n_{\mathrm{port}} = 12,$$

and the reversible write/verify register gives the oriented count

$$m_{\mathrm{rep}} = 2\dim\big(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)\big) = 2(8+3+1) = 24.$$

These are finite-register counts. A worldsheet central charge counts gapless conformal degrees of freedom, weighted by field type. For example, twenty-four chiral bosons have $c = 24$, but twenty-four chiral Majorana fermions have $c = 12$, and massive fields do not contribute to the infrared central charge [18, 19].

The heat-kernel identity also cannot determine c_L and c_R by itself. One can tensor the sewn edge sector with a spectator CFT without changing

$$Z_{\mathrm{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)} = K_t(1),$$

while shifting the total central charge. Therefore port counting and the heat-kernel trace feed a criticality test. Criticality requires the finite-carrier critical-edge certificate below.

Theorem 9.1 (Boundary from OPH edge data)**.** *The OPH edge statements used here do not imply*

$$\operatorname{rank} k_L = 24, \qquad \operatorname{rank} k_R = 12, \qquad T_{\mathrm{OPH}} = T_{\mathrm{Sug}}, \qquad G^2 \sim T_R.$$

Indeed the rank equations are not invariantly meaningful until a continuum current algebra and its level matrices k_L, k_R have been specified.

Proof. There are four independent obstructions.

First, the twelve curvature ports are not themselves twelve independent currents. Within the declared integer-counting unit-defect realization, define defect fluctuations q_a, $a = 1, \ldots, 12$. Their total charge is fixed:

$$\sum_{a=1}^{12} q_a = 12.$$

For currents $J_a = \partial q_a$, this gives

$$\sum_{a=1}^{12} J_a = 0, \qquad \sum_a k_{ab} = 0,$$

so the all-ones vector lies in the kernel of the current two-point matrix and

$$\operatorname{rank} k \leq 11.$$

On the exact unit-defect normal form, $q_a = 1$ for every port, these curvature currents vanish. Thus the curvature-defect variables cannot be the required twelve independent chiral fields; any such fields must be additional dynamical variables living at the ports.

The half-unit display rescales the signed event values and whole-unit repair threshold together. It is a change of units on the same move graph and therefore supplies neither a second counting normalization nor an independent candidate. The central-charge obstruction above is exact inside the declared counting realization.

Second, orientation labels do not by themselves double the rank. If verification is the reverse of writing, $J_{a,-} = -J_{a,+}$, and the unoriented covariance is K, then

$$K_{\text{orient}} = \begin{pmatrix} K & -K \\ -K & K \end{pmatrix} \sim \begin{pmatrix} 0 & 0 \\ 0 & 2K \end{pmatrix}.$$

Hence

$$\operatorname{rank} K_{\text{orient}} = \operatorname{rank} K,$$

not twice that rank. Doubling would require an independent covariance block

$$K_{\text{independent}} = \begin{pmatrix} K & 0 \\ 0 & K \end{pmatrix}.$$

Both covariance structures are compatible with two operation labels. OPH counts write/verify operations; independent-field covariance is a separate certificate.

Third, twelve Standard Model generators are not automatically central charge 12. For a non-abelian affine algebra $\widehat{\mathfrak{g}}_k$, the Sugawara central charge is

$$c(\mathfrak{g}_k) = \frac{k \dim \mathfrak{g}}{k + h^\vee}$$

[19]. Interpreting the Standard Model algebra as level-one affine currents gives

$$\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_1 \oplus \mathfrak{su}(2)_1 \oplus \mathfrak{u}(1),$$

and hence

$$c = \frac{8}{1+3} + \frac{3}{1+2} + 1 = 2 + 1 + 1 = 4,$$

not 12. Two independent copies would give $c = 8$, not 24. To obtain $c = 12$ or $c = 24$, the edge slots must be proved equivalent to independent Heisenberg/free-boson currents, or to another CFT with those central charges.

Fourth, a current algebra can be a proper subsector of a larger CFT:

$$T_{\text{OPH}} = T_{\text{Sug}} + T_{\text{coset}}, \qquad c_{\text{OPH}} = c_{\text{Sug}} + c_{\text{coset}}.$$

The equality $T_{\text{OPH}} = T_{\text{Sug}}$ requires $T_{\text{coset}} = 0$. Neither the twelve-port theorem nor $Z_{\text{edge}} = K_t(1)$ proves this. Likewise, a worldsheet supercurrent must be a local, fermion-odd field of conformal weight 3/2 satisfying the super-Virasoro OPE. A square root of a discrete repair operation is not enough; one needs a $\mathbb{Z}_2$ grading, fermionic locality, spin structures, and the supercurrent OPE. □

The precise claim is conditional: a heterotic edge-CFT completion would make the ranks, stress-tensor equality, supercurrent, and critical dimension follow rigidly.

Definition 9.2 (Critical-edge finite-carrier receipts)**.** *Let V_{port} be the real vector space of dynamical port-amplitude perturbations, distinct from the fixed curvature charges $q_a = 1$, and let*

$$\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1), \qquad \dim \mathfrak{g}_{\text{SM}} = 12.$$

The finite-carrier critical-edge certificate consists of five receipts.

1. *A visible-response map*

$$\mathcal{R}_{\rm port} : \mathfrak{g}_{\rm SM} \to V_{\rm port}$$

with

$$\operatorname{rank} \mathcal{R}_{\rm port} = 12.$$

2. *Positive covariance levels in every icosahedral and oriented sector:*

$$\kappa_{\mathbf{1}}, \kappa_{\mathbf{3}}, \kappa_{\mathbf{3}'}, \kappa_{\mathbf{5}} > 0, \qquad \kappa_{\rho,\pm} > 0$$

for $\rho \in \{\mathbf{1}, \mathbf{3}, \mathbf{3}', \mathbf{5}\}$.

3. *Scaled additive edge records whose continuum OPE is abelian Heisenberg:*

$$J_A(z)J_B(w) \sim \frac{k_{AB}}{(z-w)^2}.$$

4. *A local graded right-moving half-repair operator* Q_N *such that*

$$\{Q_N, (-1)^F\} = 0, \qquad Q_N^2 - H_{R,N} \to 0$$

in the refinement limit.

5. *A torus and spin-structure receipt*

$$Z_{\rm OPH}(\tau, \bar\tau) = q^{-1}\bar q^{-1/2}(1 + \cdots)$$

with the modular transformations of the heterotic spin-structure sum.

9.1 Operational finite-carrier receipt protocol

The finite-carrier certificate is an executable protocol, not a synonym for a single twelve-port cavity. A single Echosahedron is a recurrent zero-spatial-dimensional patch: it evolves and self-reads, but the spatial edge coordinate needed for a (1+1)-dimensional scaling theory is absent. Central charge, current modes, and Virasoro closure require a sewn edge cylinder

$$\text{edge cylinder}(L), \qquad \mathcal{H}_L = \bigotimes_{x=0}^{L-1} \mathcal{H}_{{\rm cell},x}, \qquad x \equiv x + L.$$

The local update on this cylinder must be inherited from the OPH read–compare–repair–commit cycle, not chosen to reproduce a desired CFT. Its exported object is a transfer operator

$$T_L = e^{-aH_L},$$

or, for a stochastic carrier, a Markov/Koopman transfer operator with a reflected-positive Euclidean interpretation.

The operational receipt suite is:

FC-1. Carrier and refinement manifest. Export model hashes, local factor dimensions, update/repair gates, transfer callbacks, the action of the alternating group A_5 on five letters, orientation involution, translation operator, record/port operators, refinement maps, precision model, and either a positive self-adjoint transfer matrix or reflected-positive Gram matrices.

FC-2. Twelve-port dynamical rank. Perturb the twelve ports around settled states, measure the full response matrix, decompose the A_5 representation $\mathbb{R}^{12} \cong \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$, and require all irreducible response coefficients to remain nonzero under refinement.

FC-3. Orientation independence. Form $V_{\text{port}} \otimes \mathbb{R}[C_2]$, project to the $A_5 \times C_2$ sectors, and verify that both orientation parities survive dynamically. This is the immediate falsifier for mere write/verify duplication.

FC-4. Conserved currents. Discover local densities and currents satisfying the discrete continuity equation on the edge cylinder, then measure chiral level matrices whose stable ranks approach

$$\operatorname{rank} k_L = 24, \qquad \operatorname{rank} k_R = 12.$$

FC-5. Virasoro central charge. Construct lattice stress-tensor modes and fit the Virasoro residuals on low-energy projectors, with independent finite-size energy checks, targeting

$$c_L \to 24, \qquad c_R \to 12.$$

FC-6. Sugawara exhaustion. Build the Sugawara stress tensor from the measured currents and require $T_{\text{OPH}} - T_{\text{Sug}} \to 0$, equivalently no residual positive-central-charge coset.

FC-7. Supercurrent. Exhibit a genuine right-moving graded sector and a local odd supercurrent whose finite-carrier square-root receipt satisfies $Q_N^2 - H_{R,N} \to 0$ on the low-energy projector.

FC-8. Torus, spin structures, and internal lattice. Export spin-structure partition functions, verify their modular transformation law after the full GSO projection, and identify the rank-sixteen even self-dual left-moving internal lattice. The Bouchard-Donagi witness belongs to the $E_8 \oplus E_8$ heterotic branch.

FC-9. Blind refinement and falsification. Pre-register several (L, a) refinements, report all failures, and run controls that collapse orientation, permute ports, break icosahedral geometry, add spectators or masses, reverse chirality conventions, and disconnect seams.

The existing twelve-port calibration model can test FC-0, FC-1, and the precursor part of FC-2. It cannot by itself test $\operatorname{rank} k_L = 24$, $\operatorname{rank} k_R = 12$, $T_{\text{OPH}} = T_{\text{Sug}}$, the right-moving supercurrent, or $(c_L, c_R) = (24, 12)$. Those require the edge-cylinder extension. Without such receipts, the heterotic central-charge identification is not certified, not a passed numerical result.

Lemma 9.3 (Port-spectrum and icosahedral rank tests)**.** *If every nonzero observer-visible repair generator changes some dynamical port record, then $\mathcal{R}_{\text{port}}$ is an isomorphism. If the twelve ports transform as the vertex permutation representation of A_5, then*

$$\mathbb{R}^{12} \cong \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

For an A_5-invariant right-moving covariance matrix,

$$k_R = \kappa_{\mathbf{1}} P_{\mathbf{1}} + \kappa_{\mathbf{3}} P_{\mathbf{3}} + \kappa_{\mathbf{3}'} P_{\mathbf{3}'} + \kappa_{\mathbf{5}} P_{\mathbf{5}},$$

and

$$\operatorname{rank} k_R = 12 \quad \Longleftrightarrow \quad \kappa_{\mathbf{1}}, \kappa_{\mathbf{3}}, \kappa_{\mathbf{3}'}, \kappa_{\mathbf{5}} > 0.$$

For the oriented space $V_{\text{port}} \otimes \mathbb{R}[C_2]$,

$$\operatorname{rank} k_L = 24 \quad \Longleftrightarrow \quad \kappa_{\rho,\pm} > 0 \quad \textit{for all eight oriented sectors.}$$

Proof. The response-map condition says $\ker \mathcal{R}_{\mathrm{port}} = 0$. Since the domain and codomain both have dimension 12, $\mathcal{R}_{\mathrm{port}}$ is invertible. The A_5 permutation character on the twelve icosahedral vertices is

$$\chi_V = (12, 0, 0, 2, 2),$$

which decomposes as

$$\chi_{\mathbf{1}} + \chi_{\mathbf{3}} + \chi_{\mathbf{3'}} + \chi_{\mathbf{5}} = (12, 0, 0, 2, 2).$$

Schur's lemma then diagonalizes any A_5-invariant covariance by irreducible sector. Reflection positivity makes nonzero sector coefficients positive. The oriented statement is the same argument applied to the $A_5 \times C_2$ decomposition. □

Lemma 9.4 (Observability and current emergence)**.** *If the current covariance is positive semidefinite, all invisible directions have been quotiented out, and every remaining nonzero port perturbation changes an observer-visible record statistic, then the current covariance is positive definite. If additive port records satisfy*

$$Q_A(I_1 \cup I_2) = Q_A(I_1) + Q_A(I_2)$$

and are conserved under accepted repairs away from interval endpoints, then in a local unitary scale-invariant sewn-edge limit they become chiral currents

$$Q_A(I) = \int_I J_A(z)\, dz, \qquad \bar{\partial} J_A = 0.$$

If the fixed-cutoff integrated charges commute, their continuum current algebra is abelian:

$$J_A(z) J_B(w) \sim \frac{k_{AB}}{(z-w)^2}.$$

Proof. For a positive-semidefinite covariance, a nonzero vector v with $v^T k v = 0$ defines a zero-norm combination $J_v = v^A J_A$, hence an invisible fluctuation in the observer-facing algebra. The quotient-observability hypotheses exclude such a vector. Additivity gives a local continuum density, conservation gives a conserved two-dimensional current, and chirality gives a holomorphic component of weight $(1, 0)$. Commuting zero modes exclude a simple-pole affine term, leaving only the central $(z-w)^{-2}$ singularity. □

Theorem 9.5 (Critical-edge certificate theorem)**.** *If the five finite-carrier receipts above hold, then the OPH sewn-edge scaling branch satisfies the heterotic edge-polarization hypothesis below. Consequently the physical edge CFT has*

$$(c_L^{\mathrm{phys}}, c_R^{\mathrm{phys}}) = (24, 12),$$

a right-moving $N = 1$ supercurrent, no residual coset, and an even self-dual rank-sixteen left-moving internal lattice.

Proof. The rank and observability receipts make the right-moving port covariance positive definite of rank 12 and the oriented left-moving covariance positive definite of rank 24. The current-emergence receipt turns the additive records into abelian Heisenberg currents. The abelian Sugawara construction gives central charge equal to rank, hence $c_R = 12$ and $c_L = 24$ for the generated sectors.

The torus vacuum exponent

$$q^{-1}\bar{q}^{-1/2} = q^{-c_L/24}\bar{q}^{-c_R/24}$$

fixes the total central charges to the same values. Therefore the residual stress tensor

$$T_{\text{res}} = T_{\text{OPH}} - T_{\text{Sug}}$$

has $c_{\text{res}} = 0$ in both chiralities. In a unitary positive-energy CFT with unique vacuum, a $c = 0$ Virasoro sector is null after quotienting null states, so there is no residual coset.

The local graded half-repair operator supplies the refinement limit of a right-moving odd square root of translation. Its density is the supercurrent G_R, and the free $N = 1$ OPE gives

$$G_R(\bar{z})G_R(\bar{w}) \sim \frac{2c_R/3}{(\bar{z}-\bar{w})^3} + \frac{2T_R(\bar{w})}{\bar{z}-\bar{w}}.$$

The modular spin-structure receipt then identifies the sixteen remaining left-moving units with an even self-dual rank-sixteen lattice sector. □

Assumption 9.6 (Heterotic edge-polarization hypothesis)**.** *The physical sewn-edge scaling limit is isomorphic, as a graded chiral CFT, to*

$$\text{ScaleLim}(\textit{OPH sewn-edge carrier}) \cong (\text{Heis}_8 \otimes V_{\Gamma_{16}})_L \otimes (\text{Heis}_8 \otimes \text{Ferm}_8)_R\,,$$

with no residual coset or spectator CFT. Here Heis_8 *is generated by eight transverse free bosons,* $V_{\Gamma_{16}}$ *is a rank-sixteen even self-dual lattice VOA on the left-moving side, and* Ferm_8 *is generated by eight right-moving Majorana fermions with a consistent spin-structure projection.*

Theorem 9.7 (Heterotic completion theorem)**.** *Under Assumption 9.6, the physical chiral central charges are*

$$\boxed{c_L^{\text{phys}} = 24, \qquad c_R^{\text{phys}} = 12,}$$

the relevant Heisenberg level matrices have ranks 24 *and* 12*, the stress tensors are Sugawara stress tensors for those Heisenberg currents, and the right-moving sector carries an* $N = 1$ *supercurrent. The shared target spacetime dimension is then forced to be* $D = 10$*, with left-moving internal central charge* 16*.*

Proof. On the left, write

$$\Phi_L^A = (X_L^1, \ldots, X_L^8, Y_L^1, \ldots, Y_L^{16}), \qquad A = 1, \ldots, 24,$$

with

$$\Phi_L^A(z)\Phi_L^B(w) \sim -\delta^{AB}\log(z-w).$$

The currents $J_L^A = i\partial\Phi_L^A$ obey

$$J_L^A(z)J_L^B(w) \sim \frac{\delta^{AB}}{(z-w)^2},$$

so $k_L = I_{24}$ and rank $k_L = 24$. The abelian Sugawara tensor

$$T_L = \frac{1}{2}\sum_{A=1}^{24} : J_L^A J_L^A :$$

has Virasoro central charge $c_L = 24$, and by the no-coset clause it is the full left-moving stress tensor.

On the right, take eight transverse bosons X_R^i and eight Majorana fermions ψ^i:

$$X_R^i(\bar z)X_R^j(\bar w) \sim -\delta^{ij}\log(\bar z-\bar w), \qquad \psi^i(\bar z)\psi^j(\bar w) \sim \frac{\delta^{ij}}{\bar z-\bar w}.$$

Pairwise bosonization of the fermions,

$$\frac{\psi^{2a-1}\pm i\psi^{2a}}{\sqrt{2}} = e^{\pm iH^a}, \qquad a=1,\ldots,4,$$

shows that their stress tensor is equivalent to four chiral bosons. A Heisenberg-current basis is

$$J_R^\alpha = \left(i\bar\partial X_R^1,\ldots,i\bar\partial X_R^8, i\bar\partial H^1,\ldots,i\bar\partial H^4\right), \qquad \alpha=1,\ldots,12,$$

with level matrix $k_R = I_{12}$. Thus $\operatorname{rank} k_R = 12$, and the right-moving stress tensor is the full abelian Sugawara tensor in the bosonized basis:

$$T_R = \frac{1}{2}\sum_{\alpha=1}^{12} :J_R^\alpha J_R^\alpha:, \qquad c_R = 8+\frac{8}{2} = 12.$$

The unbosonized expression gives the standard $N=1$ supercurrent

$$G_R(\bar z) = i\sum_{i=1}^{8}\psi^i(\bar z)\bar\partial X_R^i(\bar z).$$

Wick contraction yields

$$G_R(\bar z)G_R(\bar w) \sim \frac{8}{(\bar z-\bar w)^3} + \frac{2T_R(\bar w)}{\bar z-\bar w}.$$

Since $2c_R/3 = 2(12)/3 = 8$, this is precisely the $N=1$ super-Virasoro OPE, so

$$G^2 \sim T_R.$$

Let $n = D-2$ be the number of physical transverse spacetime coordinates. On the supersymmetric chirality,

$$c_R^{\text{phys}} = n+\frac{n}{2} = \frac{3}{2}n.$$

Setting $c_R^{\text{phys}} = 12$ gives $n=8$, hence

$$\boxed{D=10.}$$

On the bosonic chirality,

$$c_L^{\text{phys}} = n + c_{\text{internal}} = 24,$$

so

$$\boxed{c_{\text{internal}} = 16.}$$

Covariantly,

$$c_L^{\text{matter}} = 10+16 = 26, \qquad c_R^{\text{matter}} = 10+\frac{10}{2} = 15,$$

the matter central charges canceled by the bosonic ghosts on the left and the supersymmetric ghost system on the right. □

Theorem 9.8 (Modular lattice and anomaly closure)**.** *The rank-sixteen internal lattice in Assumption 9.6 must be even and self-dual. Hence*

$$\Gamma_{16} = E_8 \oplus E_8 \quad \textit{or} \quad \Gamma_{16} = D_{16}^+,$$

corresponding to the heterotic gauge groups $E_8 \times E_8$ and $\mathrm{Spin}(32)/\mathbb{Z}_2$. The Bouchard–Donagi construction lies on the $E_8 \oplus E_8$ branch. The lattice theorem does not select that branch over D_{16}^+, and compactification dualities require a separate equivalence check. With $D = 10$, the covariant Weyl anomalies cancel:

$$(10 + 16) - 26 = 0, \qquad \left(10 + \frac{10}{2}\right) - 15 = 0.$$

Proof. For the internal bosons,

$$Z_{\Gamma_{16}}(\tau) = \frac{\Theta_{\Gamma_{16}}(\tau)}{\eta(\tau)^{16}}, \qquad \Theta_\Gamma(\tau) = \sum_{p\in\Gamma} q^{p^2/2}.$$

Invariance under $T : \tau \mapsto \tau+1$ requires $p^2 \in 2\mathbb{Z}$, so the lattice is even. Under $S : \tau \mapsto -1/\tau$, Poisson resummation maps Γ to its dual Γ^*, so modular closure requires $\Gamma^* = \Gamma$. The positive-definite even unimodular rank-sixteen lattices are $E_8 \oplus E_8$ and D_{16}^+. The ghost central charges are -26 on the bosonic left and -15 on the supersymmetric right, giving the displayed anomaly cancellations. BRST nilpotency follows with the standard intercept and level-matching conditions. ☐

The OPH theorem target is precise:

$$\boxed{\mathrm{ScaleLim}(\text{OPH sewn-edge carrier}) \cong (\mathrm{Heis}_8 \otimes V_{\Gamma_{16}})_L \otimes (\mathrm{Heis}_8 \otimes \mathrm{Ferm}_8)_R}$$

with no residual coset. It cannot be replaced by the numerical observation $12 \to 24$. It must be established by constructing the edge transfer operator, identifying the gapless chiral fields, calculating their OPEs, deriving the fermionic grading and spin-structure sum, and proving stress exhaustion. Equivalently, the finite carrier must export the five receipts listed above:

$$\operatorname{rank} \mathcal{R}_{\text{port}} = 12, \quad \kappa_\rho > 0, \quad \kappa_{\rho,\pm} > 0, \quad J_A J_B \sim k_{AB}(z-w)^{-2}, \quad Q_N^2 - H_{R,N} \to 0,$$

together with the torus receipt $Z_{\text{OPH}} = q^{-1}\bar{q}^{-1/2}(1 + \cdots)$ and the correct modular spin-structure transformations. Once that edge-VOA identification is proved, $D = 10$ and the left-moving $c_{\text{internal}} = 16$ sector follow by the calculation above. The number 26 belongs to the left-moving bosonic matter central charge, not to twenty-six shared heterotic spacetime dimensions.

10 The OPH Tensor Carrier and the String Graviton

The tensor dictionary is the conceptual hinge. OPH supplies an action-level classical transverse-traceless metric carrier on the stated pure-Einstein branch. The critical-string candidate must separately supply a physical quantum state and pole and show that its classical limit is that same carrier.

10.1 OPH side: metric from modular geometry

On the OPH gravity branch, cap modular flow becomes geometric in the support-visible scaling limit, and fixed-cap generalized-entropy stationarity supplies the Einstein relation. The metric is the compressed observer-facing datum that makes the cap modular action geometric. A small metric perturbation is a deformation of the OPH geometric normal form:

$$q_{ab}^{\mathrm{OPH}} \longmapsto q_{ab}^{\mathrm{OPH}} + h_{ab}^{\mathrm{OPH}}.$$

The transverse-traceless part

$$h_{ab}^{\mathrm{OPH,TT}}$$

is the propagating classical spin-two perturbation. This statement alone does not construct a graviton Hilbert space or a particle pole. Those are additional receipts demanded from the critical-string representative below.

10.2 String side: metric from the closed-string vertex

In a critical closed string, the metric appears as a background coupling in the sigma model:

$$S_\sigma \supset \frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{\gamma}\, \gamma^{\alpha\beta} G_{\mu\nu}(X) \partial_\alpha X^\mu \partial_\beta X^\nu.$$

Perturbing

$$G_{\mu\nu} = G_{\mu\nu}^{(0)} + \kappa h_{\mu\nu}$$

produces the standard closed-string spin-two vertex

$$V_h(k, \epsilon) = \epsilon_{\mu\nu}\, \partial X^\mu \bar{\partial} X^\nu e^{ik\cdot X}.$$

Equivalently, at the first closed-string oscillator level,

$$|G; k, \epsilon\rangle = \epsilon_{\mu\nu}^{\mathrm{TT}} \alpha_{-1}^\mu \tilde{\alpha}_{-1}^\nu |0; k\rangle.$$

The polarization decomposes into symmetric traceless, antisymmetric, and trace pieces:

$$\epsilon_{\mu\nu} = \epsilon_{(\mu\nu)}^{\mathrm{TT}} + \epsilon_{[\mu\nu]} + \frac{1}{D} \eta_{\mu\nu} \epsilon^\lambda{}_\lambda.$$

The symmetric transverse-traceless piece is the string graviton. The antisymmetric piece is the B-field, and the trace is the dilaton.

10.3 Left-right collar factorization

A closed OPH edge cycle has two collar readings. The two readings are sewn into the heat-kernel factor d_R^2. In the large-edge worldsheet language they become the left and right chiral oscillator sectors:

$$\text{left collar edge mode} \otimes \text{right collar edge mode} \longmapsto \alpha_{-1}^\mu \tilde{\alpha}_{-1}^\nu |0; k\rangle.$$

The level-one tensor product splits after contraction with a polarization tensor. A general polarized state decomposes as

$$\epsilon_{\mu\nu} \alpha_{-1}^\mu \tilde{\alpha}_{-1}^\nu |0; k\rangle = \epsilon_{(\mu\nu)}^{\mathrm{TT}} \alpha_{-1}^\mu \tilde{\alpha}_{-1}^\nu |0; k\rangle + \epsilon_{[\mu\nu]} \alpha_{-1}^\mu \tilde{\alpha}_{-1}^\nu |0; k\rangle + \frac{1}{D} \eta_{\mu\nu} \epsilon^\lambda{}_\lambda \alpha_{-1}^\mu \tilde{\alpha}_{-1}^\nu |0; k\rangle.$$

The proposed quantum completion of the OPH tensor carrier is the symmetric transverse-traceless component of the two-sided collar excitation, read in the critical worldsheet language:

$$h_{\mu\nu}^{\mathrm{OPH,TT}} \longleftrightarrow \epsilon_{\mu\nu}^{\mathrm{TT}} \alpha_{-1}^\mu \tilde{\alpha}_{-1}^\nu |0; k\rangle.$$

10.4 The map

Define the OPH-carrier-to-string-state map by

$$\mathfrak{M}_{\text{grav}} : [h_{ab}^{\text{OPH,TT}}] \longmapsto [\epsilon_{\mu\nu}^{\text{TT}} \partial X^\mu \bar{\partial} X^\nu e^{ik\cdot X}],$$

with the following normalization condition:

$$G_4^{\text{string}}(m_\star, \alpha', g_s, V_6) = G_4^{\text{OPH}}(N_{\text{scr}}, P).$$

OPH fixes the classical Einstein-frame tensor normalization. The string representative must supply the critical worldsheet realization, BRST-physical state space, positive norm, and massless pole.

Theorem 10.1 (Conditional tensor-carrier quantum completion)**.** *Suppose a critical-string presentation satisfies all of the following: (i) the OPH pure-Einstein flat-background quadratic receipt with two transverse-traceless null modes; (ii) a BRST-physical, positive-norm closed-string symmetric spin-two state with a positive-residue pole at $k^2 = 0$; (iii) equality of the four-dimensional normalization and classical background deformation; and (iv) the selector condition that no second independent visible spin-two sector survives. Then the string state is a quantum completion of the OPH classical metric carrier:*

$$\mathfrak{M}_{\text{grav}}(h^{\text{OPH,TT}}) = h^{\text{string,TT}}.$$

The absence of an additional visible spin-two particle is condition (iv) of the selector receipt, not a consequence of the classical Einstein equation alone.

Proof. The OPH gravity branch produces a single observer-facing Einstein-frame metric on the physical quotient. Under (i), its reduced quadratic action and Hamiltonian have two classical null tensor modes. Assumption (ii) places the displayed string vertex in the physical Hilbert space and gives it the required particle pole; this is precisely the information absent from a classical Einstein equation. By (iii), the vertex deforms the same normalized metric and has the OPH tensor mode as its classical limit, so the displayed map identifies one classical carrier with one quantum state. Condition (iv) rejects any candidate with an additional independent visible spin-two state. Thus the particle statement follows from all four receipts together, without identifying the classical field equation by itself. □

OPH object	**String-theory image**
Cap modular geometry	Target-space metric background $G_{\mu\nu}(X)$.
Fixed-cap generalized entropy stationarity	Low-energy Einstein equation / vanishing metric beta function on the selected background.
Metric perturbation h_{ab}^{OPH}	Closed-string background deformation $h_{\mu\nu}\partial X^\mu \bar{\partial} X^\nu$.
OPH classical tensor carrier	BRST-physical symmetric transverse-traceless closed-string state, conditional on its positive-residue pole receipt.
Screen-capacity normalization	Four-dimensional Newton constant after compactification.
Edge record/collar sewing	Worldsheet sewing and genus expansion in the controlled large-edge branch.

11 OPH-Augmented String Theory

OPH supplies a selection and readout functor for critical string vacua. Ordinary string theory supplies a large class of critical presentations. OPH asks which presentation is the critical worldsheet completion of the observer-visible edge normal form.

Definition 11.1 (OPH augmentation of a string vacuum)**.** *A string vacuum v is augmented by OPH data when it is equipped with a map*

$$\Pi_{\mathrm{OPH}}(v) = (G_{\mathrm{phys}}, Y, N_c, N_g, n_H, \mathcal{O}_{\mathrm{op}}, G_4, \Lambda, \mathcal{O}_{\mathrm{quant}})$$

from compactification data to observer-visible records, together with the constraints

$$\mathcal{C}_v(m_\star) = 0, \qquad \Pi_{\mathrm{OPH}}(v) = \Pi^{\mathrm{OPH}}_{\mathrm{target}}, \qquad \ker D\mathcal{C}_v \cap \ker D\Pi_{\mathrm{OPH}} = T(G_{\mathrm{inv}} \cdot m_\star),$$

where N_{OPH} is a physical slice transverse to independently proved OPH-invisible redundancies, as formalized in Definition 16.1. It is not defined by $\ker D\Pi_{\mathrm{OPH}}$. The constraints require a separate dynamical stability receipt. The joint kernel condition is the coordinate-free moduli-locking certificate of Theorem 13.10.

The augmented vacuum equations are:

$$\begin{cases} \beta^G = \beta^B = \beta^\Phi = 0, & \text{worldsheet consistency,} \\ (c_L^{\mathrm{phys}}, c_R^{\mathrm{phys}}) = (24, 12), \quad G^2 \sim T_R, & \text{critical-edge certificate,} \\ F^{0,2} = 0, \quad J^2 \wedge F = 0, & \text{heterotic bundle stability/HYM,} \\ c_2(TX) - c_2(V_{\mathrm{vis}}) - c_2(V_{\mathrm{hid}}) = [W], & \text{Bianchi/anomaly condition,} \\ \chi(V_{\mathrm{matter}}) = 3, \quad n_H = 1, \quad N_{\mathrm{exotic}} = 0, & \text{visible cohomology target,} \\ G_{\mathrm{phys}} = (\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6, & \text{global group target,} \\ \mathcal{O}_{\mathrm{op}} = \mathbb{Z}_4^R \text{ safety}, & \text{operator target,} \\ G_4^{\mathrm{string}} = G_4^{\mathrm{OPH}}, \quad \alpha' \mapsto \alpha'_{\mathrm{OPH}}, & \text{graviton normalization target,} \\ \nabla V_{\mathrm{eff}}(m_\star) = 0, \quad \mathrm{Hess}_{\mathrm{phys}}\, V_{\mathrm{eff}} \succeq \mu^2 I, & \text{dynamical stabilization,} \\ \mathcal{H}_v(m_\star) = 0, \quad \operatorname{rank} D\mathcal{H}_v = \dim \mathcal{X}_v, & \text{constraint-augmented target locking.} \end{cases}$$

For a finite catalogue with certified parameter-domain coverage, these equations define a finite certificate problem. A conventional string vacuum can be internally consistent and fail the OPH observer-visible target map. No finiteness or exhaustiveness claim follows for the unrestricted candidate universe.

11.1 Falsifiable Consequences of a Certified Witness Class

A certified witness class has the following falsifiable consequences:

1. **One visible massless Higgs pair at the compactification target.** Extra massless Higgs pairs fail the OPH one-Higgs electroweak branch.

2. **No massless chiral exotics in the published cohomology table.** Exotic chiral matter changes the observer-visible matter package and fails the cohomology target.

3. **No extra visible low-scale** U(1)**.** Additional visible abelian gauge bosons change the OPH global group quotient.

4. **No mixed X/Y generator in the connected visible adjoint.** The selected visible group is the product quotient. A propagating connection component requires an action receipt, while Wilson-line and Kaluza–Klein particle modes require the separate spectrum and coupling calculation.

5. **Operator-safety pattern.** Perturbative R-parity violation, perturbative dimension-five proton decay, and a perturbative μ-term are forbidden on the operator-safe branch; Yukawas and the Weinberg neutrino operator are allowed.

6. **One visible spin-two sector.** Any additional physical spin-two pole, bimetric sector, or independent low-energy tensor polarization contradicts selector uniqueness; the classical Einstein equation alone does not establish this exclusion.

12 String-Community Pressure Points

The string-community problem is selection. The 2024 Standard Model review states the challenge in compactification language: reproduce the Standard Model gauge sector, chiral matter, and phenomenology from geometry and topology [26]. Heterotic M-theory stabilization work emphasizes the computational weight of dilaton, complex-structure, and Kähler moduli stabilization [27]. The philosophy literature on the string landscape frames the predictivity concern as underdetermination among many solutions compatible with observed data [28]. Exact flux-vacua work also makes the computational problem concrete: stabilized flux vacua require solving vacuum conditions with exponential corrections, although symmetry loci can turn some conditions algebraic [29].

OPH imposes the observer-visible target

$$\mathfrak{S}_{\mathrm{OPH}} = \begin{pmatrix} \dfrac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \quad N_c = 3, \quad N_g = 3, \quad Y_{\mathrm{SM}}, \\ n_H = 1, \quad \text{no light chiral exotics}, \quad \mathbb{Z}_4^R \text{ safety} \end{pmatrix}.$$

The named geometric structural row in the package is

$$\boxed{BD^{\mathrm{OPH}}_{n=1} = \text{Bouchard-Donagi } E_8\times E_8 \text{ heterotic SU(5) Standard Model, one-Higgs-pair stratum.}}$$

The tested operator-safe proposal is

$$\boxed{BD^{\mathrm{OPH}}_{n=1,+} = BD^{\mathrm{OPH}}_{n=1} + \mathbb{Z}_4^R.}$$

Bouchard and Donagi introduced a heterotic Standard Model with the visible massless cohomology spectrum of the MSSM, no massless visible exotics, observable $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$, a Calabi–Yau threefold with $\mathbb{Z}_2$ fundamental group, and an invariant SU(5) bundle. Depending on moduli, the model has zero, one, or two Higgs doublet conjugate pairs; the one-pair region gives precisely the massless MSSM Higgs content [20]. This cohomology calculation determines zero modes, rather than the nonzero Dirac/Laplacian, Kaluza–Klein, winding, oscillator, Wilson-line, vectorlike, five-brane, or hidden spectrum [23].

The global group identity is central:

$$\mathrm{Cent}_{\mathrm{SU}(5)}(\mathrm{diag}(1,1,1,-1,-1)) = S(\mathrm{U}(3)\times\mathrm{U}(2)),$$

$$S(\mathrm{U}(3)\times\mathrm{U}(2)) \cong \frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}.$$

This fixes the charge lattice, beyond Lie-algebra matching.

Theorem 12.1 (OPH vacuum sieve theorem)**.** *A critical string vacuum $\mathcal{T}$ is OPH-admissible only if*

$$\Pi_{\mathrm{IR}}^{\mathrm{OPH}}(\mathcal{T}) = \mathfrak{S}_{\mathrm{OPH}}.$$

For the package studied here, the named geometric structural row is $BD_{n=1}^{\mathrm{OPH}}$, and the operator-safe structural row is $BD_{n=1,+}^{\mathrm{OPH}}$. This theorem is a necessary sieve statement; it does not override the failed rank condition in Corollary 3.5.

Theorem 12.2 (Global group locking theorem)**.** *The OPH visible group and the Bouchard-Donagi Wilson-line centralizer agree:*

$$G_{\mathrm{phys}} = S(\mathrm{U}(3) \times \mathrm{U}(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

For $\rho(-1) = \mathrm{diag}(1, 1, 1, -1, -1)$,

$$\mathrm{Cent}_{\mathrm{SU}(5)}(\rho) = S(\mathrm{U}(3) \times \mathrm{U}(2)).$$

Theorem 12.3 (BD structural geometry witness theorem)**.** *The Bouchard-Donagi $\mathbb{Z}_2$-quotient* SU(5)*-bundle construction supplies a nonempty heterotic* SU(5)*-corridor witness whose Wilson-line centralizer is the OPH global Standard Model group and whose visible massless cohomology contains the three-generation, no-exotic, one-Higgs branch used by the OPH selector. This statement does not supply a heavy-spectrum, threshold, or supersymmetric-sector decoupling certificate.*

Proof. Bouchard-Donagi construct the $E_8 \times E_8$ heterotic model on a Calabi-Yau quotient $X = \widetilde{X}/\mathbb{Z}_2$ with an invariant SU(5) bundle and a $\mathbb{Z}_2$ Wilson line [20]. Their bundle is built on the cover using an extension of rank-two and rank-three pieces,

$$0 \to V_2 \to \widetilde{V}^* \to V_3 \to 0, \qquad V_i = \pi'^* W_i \otimes \pi^* L_i.$$

The resulting observable zero-mode sector has the Standard Model gauge algebra, three generations, no massless visible exotic matter, and moduli regions with 0, 1, or 2 massless Higgs doublet conjugate pairs. The OPH target chooses the one-pair region. The centralizer theorem above supplies the global quotient

$$S(\mathrm{U}(3) \times \mathrm{U}(2)) \cong (\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6.$$

The BD geometry supplies the named heterotic witness and its Wilson-line centralizer. The conditional OPH matrix-current and matter packet supplies the matching maximal faithful image. It does not source-select the physical $\mathbb{Z}_6$ quotient. This sieve declares agreement of the Wilson-line centralizer with that conditional finite image and the Higgs stratum as selector inputs. Laboratory and continuum attachment are separate premises. □

13 String-Theory Questions in the OPH Sieve

The table separates exact algebraic results from conditional physical statements. Cohomology reproduction, geometric realization of the $\mathbb{Z}_4^R$ safety layer, full Yukawa computation, the nonzero-mode and hidden spectra, threshold matching, the low-energy decoupling map, and moduli locking require external certificates built from the Bouchard–Donagi geometry and its physical completion.

String-theory pressure point	OPH result	Mathematical boundary
Vacuum selection	Converts candidate vacua into a public sieve against $\mathfrak{S}_{\mathrm{OPH}}$.	Exact selector definition; a broader candidate class requires its own coverage certificate.
Global Standard Model group	Locks the finite quotient $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$ beyond Lie-algebra data.	Exact group proof.
Charge lattice and anomalies	Uses the Standard Model hypercharge lattice with anomaly cancellation.	Exact rational arithmetic.
Yukawa admissibility	One-Higgs Yukawa terms are gauge invariant.	Exact charge-sum proof.
Three generations	$\mathbb{Z}_2$-cover arithmetic gives $N_g = 3$.	Exact index arithmetic, assuming the BD c_3 input.
Higgs multiplicity	Selects $n = 1$ among BD $n = 0, 1, 2$ strata.	Selector proof.
Connected unified-gauge adjoint	The connected product-group adjoint excludes an X/Y generator. Connection and BD particle modes require separate action and spectrum assumptions.	Exact Lie-algebra proof plus external action and heavy-mode certificates.
String emergence	Sewn edge partition equals $K_t(1)$.	Exact Peter-Weyl proof.
Worldsheet criticality	The twelve-port and twenty-four oriented-register counts define only a heterotic central-charge target until the edge VOA is identified.	Non-implication theorem plus critical-edge certificate theorem; finite-carrier premises are not established here.
Vacuum stabilization	Requires stationarity, a canonically normalized physical scalar mass matrix, and a route-appropriate stability bound.	The stated BD data contain no completed effective potential, point, or Hessian enclosure.
Target locking	Requires a physical quotient, precommitted target map, and a certified constraint-augmented full-column-rank minor.	The physical-slice definition and negative rank certificate are proved; the completed constraint map, selected point, and physical BD map are absent.
Operator safety	The MSSM $\mathbb{Z}_4^R$ layer permits Yukawas and the Weinberg operator, and forbids perturbative RPV, dimension-five proton decay, and the perturbative μ-term.	Charge algebra closes; BD geometric realization is a separate premise.

Empirical tests	Assigns the named structural row a compact list of failure-prone low-energy and threshold proxy screens.	Reproducible target equations; no branch compatibility result.

Physicist pressure point	**Statement supplied by the selector**	**Logical form**
Landscape degeneracy	OPH replaces broad vacuum search with the finite target $\mathfrak{S}_{\mathrm{OPH}}$ and a public sieve.	Selector theorem.
Global Standard Model group	The Wilson-line centralizer lands on $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$.	Exact algebra.
Three families	The $\mathbb{Z}_2$-cover index relation gives $N_g = 3$ from $\lvert c_3(\widetilde{V})\rvert = 12$.	BD input plus exact arithmetic.
Higgs multiplicity	The $n = 1$ stratum is the minimal massless-cohomology stratum under the declared zero-mode score.	Selector theorem.
Exotic matter	BD supplies the no-exotics visible massless MSSM cohomology witness.	Published zero-mode geometry certificate; heavy and vectorlike modes are outside it.
Connected unified-gauge adjoint	The connected product-group adjoint excludes mixed $(3, 2, \pm 5/6)$ generators; no connection or particle-spectrum conclusion follows from this algebra alone.	Exact Lie-algebra statement.
Critical dimension	If the sewn-edge carrier has the heterotic polarization $(\mathrm{Heis}_8 \otimes V_{\Gamma_{16}})_L \otimes (\mathrm{Heis}_8 \otimes \mathrm{Ferm}_8)_R$, then $D = 10$ and $c_{\mathrm{internal}} = 16$.	Conditional edge-VOA statement.
RPV and μ-problem	$\mathbb{Z}_4^R$ permits Yukawas, forbids perturbative RPV and dimension-five proton decay, forbids perturbative μ, and leaves matter parity.	Exact charge algebra; compactification certificate required.
Vacuum stabilization	The problem requires a controlled effective action, stationarity, and a physical Hessian stability bound.	No BD completion supplies these data.
Target identifiability	The problem requires a scheme-locked physical map plus a certified constraint-augmented rank and interval-isolation receipt.	The transverse-rank obstruction of Theorem 16.7: the published pre-completion slice has 142 real directions, the five-row proxy has rank zero, and no completed constraint map, physical slice, or Jacobian is supplied.

Predictivity	Higgs/top, stop, and one-loop gauge-running coordinates give failure-prone proxy targets.	Reproducible target-side proxies only; the stated data contain no spectrum or threshold certificate.

13.1 Problem 1: vacuum selection

String theory supplies many compactifications with Standard-Model-like low-energy data. OPH adds an independent target:

$$\mathfrak{S}_{\mathrm{OPH}} = \begin{array}{c}\left(\dfrac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, N_c = 3,\, N_g = 3,\, Y_{\mathrm{SM}},\, n_H = 1\right), \\ \text{no light chiral exotics,} \qquad \text{no extra visible low-scale U(1).}\end{array}$$

Theorem 13.1 (Set-theoretic selector reduction)**.** *Let $\mathcal{C}$ be any class of critical string compactifications with an observer-visible infrared projection $\Pi_{\mathrm{IR}}^{\mathrm{OPH}}$. Define*

$$\mathcal{C}_{\mathrm{OPH}} = \{\mathcal{T} \in \mathcal{C} : \Pi_{\mathrm{IR}}^{\mathrm{OPH}}(\mathcal{T}) = \mathfrak{S}_{\mathrm{OPH}}\}.$$

The OPH selection problem over $\mathcal{C}$ compares $\mathcal{C}_{\mathrm{OPH}}$ with the quantitative target $\mathcal{O}_{\mathrm{OPH}}$. This comparison is effective and finite or enumerable only when a separate enumeration and domain-coverage certificate for $\mathcal{C}$ is supplied.

Proof. The OPH target $\mathfrak{S}_{\mathrm{OPH}}$ is fixed independently of any string compactification. The projection $\Pi_{\mathrm{IR}}^{\mathrm{OPH}}$ assigns each candidate its observer-visible group, charge lattice, matter content, Higgs count, and visible low-scale gauge factors. Equality with $\mathfrak{S}_{\mathrm{OPH}}$ is a conjunction of explicit conditions. A candidate passes exactly when all equalities and exclusions hold. The quantitative stage appends the equation

$$\mathcal{F}_{\mathcal{T}}(m) = \mathcal{O}_{\mathrm{OPH}}.$$

The selection condition is therefore a sieve and target equation. The definition alone supplies no algorithm for enumerating $\mathcal{C}$, solving every branch, or quotienting all dualities. □

13.2 Problem 2: global Standard Model group

The conditional finite matter theorem fixes the charge lattice, common kernel, and maximal faithful image. Source selection of the physical $\mathbb{Z}_6$ quotient is not claimed. A string representative passes this sieve only if it realizes the displayed maximal image and discharges that global-form attachment.

Theorem 13.2 (Centralizer computation)**.** *Let*

$$\rho = \mathrm{diag}(1, 1, 1, -1, -1) \in \mathrm{SU}(5).$$

The centralizer of ρ in SU(5) *is*

$$\mathrm{Cent}_{\mathrm{SU}(5)}(\rho) = S(\mathrm{U}(3) \times \mathrm{U}(2)).$$

Proof. The $+1$ eigenspace of ρ has dimension 3, and the -1 eigenspace has dimension 2. A unitary matrix commutes with ρ exactly when it preserves these two eigenspaces. Such a matrix is block diagonal,

$$g = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}, \qquad A \in \mathrm{U}(3), \quad B \in \mathrm{U}(2).$$

The condition $g \in \mathrm{SU}(5)$ is $\det A \det B = 1$, precisely

$$S(\mathrm{U}(3) \times \mathrm{U}(2)).$$

$\square$

Theorem 13.3 ($\mathbb{Z}_6$ quotient)**.** *There is an isomorphism*

$$S(\mathrm{U}(3) \times \mathrm{U}(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

Proof. Define

$$\varphi : \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1) \to S(\mathrm{U}(3) \times \mathrm{U}(2))$$

by

$$\varphi(A, B, z) = \begin{pmatrix} z^2 A & 0 \\ 0 & z^{-3} B \end{pmatrix}.$$

The determinant is

$$\det(z^2 A) \det(z^{-3} B) = z^6 z^{-6} \det A \det B = 1,$$

so the image lies in $S(\mathrm{U}(3) \times \mathrm{U}(2))$. The kernel consists of triples with

$$z^2 A = I_3, \qquad z^{-3} B = I_2.$$

$A = z^{-2} I_3$ and $B = z^3 I_2$. The conditions $A \in \mathrm{SU}(3)$ and $B \in \mathrm{SU}(2)$ give

$$z^{-6} = 1, \qquad z^6 = 1.$$

The kernel is

$$\{(z^{-2} I_3, z^3 I_2, z) : z^6 = 1\} \cong \mathbb{Z}_6.$$

Surjectivity follows by decomposing any $(A', B') \in S(\mathrm{U}(3) \times \mathrm{U}(2))$ into determinant-one parts and the common $\mathrm{U}(1)$ factor. The first isomorphism theorem gives the quotient. $\square$

13.3 Problem 3: charge lattice, anomalies, and one-Higgs Yukawas

The OPH target uses the Standard Model hypercharge lattice:

$$Q = (3,2)_{1/6}, \quad u^c = (\bar{3},1)_{-2/3}, \quad d^c = (\bar{3},1)_{1/3}, \quad L = (1,2)_{-1/2}, \quad e^c = (1,1)_1, \quad H = (1,2)_{1/2}.$$

Theorem 13.4 (One-generation anomaly cancellation)**.** *For one generation with the hypercharges above,*

$$\mathrm{SU}(3)^2\mathrm{U}(1) = 0, \quad \mathrm{SU}(2)^2\mathrm{U}(1) = 0, \quad \mathrm{grav}^2\mathrm{U}(1) = 0, \quad \mathrm{U}(1)^3 = 0.$$

Proof. Use left-handed Weyl fields. For $\mathrm{SU}(3)^2\mathrm{U}(1)$, the color-charged fields give

$$2 \cdot \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{2} \cdot \left(-\frac{2}{3}\right) + \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6} - \frac{1}{3} + \frac{1}{6} = 0.$$

For $\mathrm{SU}(2)^2\mathrm{U}(1)$, the weak doublets give

$$3 \cdot \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{2} \cdot \left(-\frac{1}{2}\right) = \frac{1}{4} - \frac{1}{4} = 0.$$

For the mixed gravitational anomaly, count dimensions:

$$6 \cdot \frac{1}{6} + 3 \cdot \left(-\frac{2}{3}\right) + 3 \cdot \frac{1}{3} + 2 \cdot \left(-\frac{1}{2}\right) + 1 = 1 - 2 + 1 - 1 + 1 = 0.$$

For the cubic anomaly,

$$\begin{aligned} &6\left(\frac{1}{6}\right)^3 + 3\left(-\frac{2}{3}\right)^3 + 3\left(\frac{1}{3}\right)^3 + 2\left(-\frac{1}{2}\right)^3 + 1^3 \\ &= \frac{1}{36} - \frac{8}{9} + \frac{1}{9} - \frac{1}{4} + 1 = 0. \end{aligned}$$

□

Theorem 13.5 (One-Higgs Yukawa invariants)**.** *The one-Higgs Standard Model Yukawa terms have zero hypercharge:*

$$QHu^c, \qquad QH^\dagger d^c, \qquad LH^\dagger e^c.$$

Proof. The hypercharge sums are

$$\frac{1}{6} + \frac{1}{2} - \frac{2}{3} = 0,$$
$$\frac{1}{6} - \frac{1}{2} + \frac{1}{3} = 0,$$

and

$$-\frac{1}{2} - \frac{1}{2} + 1 = 0.$$

The nonabelian indices contract using $3 \otimes \bar{3}$, $2 \otimes 2$, and the ϵ-tensor of SU(2). □

13.4 Problem 4: generation arithmetic

Bouchard-Donagi uses a $\mathbb{Z}_2$ quotient. The chiral index relation gives the number of generations:

$$N_g = \frac{|c_3(\widetilde{V})|}{2|\Gamma|}.$$

Theorem 13.6 (Three-generation cover arithmetic)**.** *If* $|c_3(\widetilde{V})| = 12$ *and* $|\Gamma| = 2$*, the quotient has three generations.*

Proof. Substitution gives

$$N_g = \frac{12}{2 \cdot 2} = 3.$$

□

13.5 Problem 5: Higgs-stratum selection

The Bouchard-Donagi construction has regions with zero, one, or two Higgs doublet conjugate pairs. The structural Higgs-count premise selects the one-pair stratum.

Theorem 13.7 (One-Higgs zero-mode minimality)**.** *Within the published Bouchard–Donagi massless-cohomology strata, the* $n = 1$ *Higgs-pair stratum is the minimal row containing a Higgs pair and no second massless Higgs pair.*

Proof. $n = 0$ lacks a Higgs pair and lacks the Yukawa-complete electroweak branch used by the OPH target. The $n = 2$ stratum has a second massless Higgs pair and therefore fails the declared zero-mode minimality score. The $n = 1$ stratum supplies exactly one massless pair. This proof does not exclude an $n = 2$ low-energy branch: moduli-dependent masses and the full decoupling calculation could lift its additional pair. □

13.6 Problem 6: connected unified-gauge adjoint

The connected product-group adjoint contains no simple-SU(5) X/Y generator. This Lie-algebra statement does not supply connection dynamics or decide whether particle modes induce proton decay.

Theorem 13.8 (Product-group adjoint)**.** *The connected adjoint of*

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}$$

is

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0).$$

It contains no $(3, 2, \pm 5/6)$ *adjoint generator; this algebraic conclusion alone does not assert a propagating connection component or particle.*

Proof. Dividing by the finite central subgroup $\mathbb{Z}_6$ leaves the Lie algebra unchanged:

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1).$$

The adjoint representation decomposes as the adjoints of the three factors:

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0).$$

Mixed $(3, 2, \pm 5/6)$ generators occur in the adjoint of simple SU(5) after breaking to the Standard Model subgroup. They are absent from the product Lie algebra above. □

13.7 Problem 7: string emergence

The worldsheet language appears from edge sewing as an effective description.

Theorem 13.9 (Heat-kernel derivation)**.** *For a compact group G, the sewn OPH edge partition*

$$Z_{\mathrm{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}$$

is the heat kernel at the identity:

$$Z_{\mathrm{edge}}(t) = K_t(1).$$

Proof. Peter-Weyl gives the heat kernel expansion

$$K_t(g) = \sum_R d_R \chi_R(g) e^{-tC_2(R)}.$$

At $g = 1$, $\chi_R(1) = d_R$. Substitution gives

$$K_t(1) = \sum_R d_R^2 e^{-tC_2(R)} = Z_{\mathrm{edge}}(t).$$

□

13.8 Problem 8: dynamical stabilization and target locking

The OPH contribution is a target criterion. This paper does not emit a completed BD moduli computation.

Theorem 13.10 (Constraint-augmented local locking)**.** *Let $\mathcal{P}$ be an n-real-dimensional ambient parameter space, let G be an independently certified OPH-invisible redundancy group with orbit dimension g on a fixed orbit-type stratum, and let*

$$\mathcal{C} : \mathcal{P} \longrightarrow \mathbb{R}^L, \qquad F : \mathcal{P} \longrightarrow \mathbb{R}^K$$

be the invariant completion constraints and a candidate-independent observer-visible readout. Suppose p satisfies $\mathcal{C}(p) = 0$, $F(p) = \mathcal{O}_{\mathrm{OPH}}$, and $D\mathcal{C}(p)$ has rank c transverse to the G-orbit. The completed quotient tangent has dimension

$$d = n - g - c.$$

The coordinate-free locking condition is

$$\ker D\mathcal{C}(p) \cap \ker DF(p) = T_p(G\cdot p).$$

It is equivalent to the following gauge-slice certificate. On a certified local slice Σ, choose c independent completion rows A and d independently registered readout rows I, and set

$$\mathcal{H}_{A,I} = (\pi_A \mathcal{C}|_\Sigma, \pi_I(F - \mathcal{O}_{\mathrm{OPH}})|_\Sigma)\,.$$

Then $\dim \Sigma = c + d$, and condition (13.10) holds exactly when some such precommitted row set has

$$\det D\mathcal{H}_{A,I}(p) \neq 0.$$

Under either form of the condition, $[p]$ is the unique nearby orbit satisfying both the completion equations and the OPH target equations.

Proof. The regular-level-set theorem gives

$$T_{[p]}(\mathcal{C}^{-1}(0)/G) \cong \ker D\mathcal{C}(p)/T_p(G\cdot p).$$

The derivative induced by F on this quotient tangent is injective exactly when its lifted kernel is the orbit tangent, which is condition (13.10). On a local slice the orbit tangent is removed. Choose c independent rows of $D\mathcal{C}$. Injectivity of DF on the d-dimensional kernel of those rows is equivalent to the existence of d readout rows making the combined $(c + d)$-square derivative invertible. The inverse-function theorem then makes the joint zero a local singleton on the slice. Quotient descent identifies that singleton with one physical orbit. □

Corollary 13.11 (BD dimension budget)**.** *Start from the documented* 142*-real-dimensional published one-Higgs quotient. If a completion adds a real physical coordinates, imposes completion equations of transverse rank r, and removes g additional independently proved invisible orbit directions, then*

$$d = 142 + a - r - g.$$

A locking certificate using k legitimate readout rows requires

$$r + g - a \geq 142 - k.$$

Thus five legitimate rows require at least 137 *net real cuts, while the three OPH rows with physical standing require at least* 139*. These bounds are necessary and not sufficient. If $d = 0$, the completion equations isolate the vacuum and no readout derivative is needed.*

Proof. The dimension formula is the regular quotient count. An injective linear map from a d-dimensional tangent space to $\mathbb{R}^k$ requires $d \leq k$. Rearranging gives the stated inequalities. $\square$

Theorem 13.12 (Dynamical stabilization certificate)**.** *Let V_{eff} be a G-invariant C^2 effective potential with a certified domain of validity. A Minkowski or de Sitter scalar vacuum is strictly locally stabilized modulo G if*

$$\nabla V_{\text{eff}}(p) = 0, \qquad \operatorname{Hess} V_{\text{eff}}(p)|_{T_{[p]}\mathcal{M}^{\text{phys}}} \succeq \mu^2 I$$

in canonically normalized physical coordinates for a certified $\mu^2 > 0$, after removing gauge and Goldstone directions. An anti-de Sitter branch instead requires the corresponding Breitenlohner–Freedman spectral bound. Full rank of an observer readout proves local identifiability; it does not give a scalar a mass. Both stabilization and target-locking receipts are required unless the declared completion equations themselves include a separately certified stability theorem.

Proof. For the Minkowski or de Sitter case, Taylor's theorem and the positive lower Hessian bound give a strict quadratic increase of V_{eff} in every sufficiently small physical displacement. The removed orbit and Goldstone directions are not physical scalar moduli. The anti-de Sitter statement uses its stability bound rather than positivity of the potential Hessian. The derivative of a separate readout does not enter either mass operator, which proves the final distinction. $\square$

Theorem 13.13 (Local isolation by a certified row minor)**.** *Let $\mathcal{F} : \mathcal{M} \to \mathbb{R}^K$ be C^1, let the physical slice have real dimension $d \leq K$, and suppose*

$$\mathcal{F}(m_\star) = \mathcal{O}_{\text{OPH}},$$

after quotienting independently proved OPH-invisible equivalences. If an explicitly identified $d \times d$ row minor of $D\mathcal{F}(m_\star)$ has nonzero determinant, then $\operatorname{rank} D\mathcal{F}(m_\star) = d$ *and $m_\star$ is locally isolated in the physical moduli directions.*

Proof. Let I be the selected set of d output rows and let π_I be the corresponding coordinate projection. The derivative of $\pi_I \circ \mathcal{F}$ is invertible at $m_\star$. The inverse-function theorem gives a neighborhood on which $\pi_I \circ \mathcal{F}$ is injective. Any point in that neighborhood with $\mathcal{F} = \mathcal{O}_{\text{OPH}}$ has the same projected value as $m_\star$, and hence equals $m_\star$. This proves local branch isolation only; it is not a global uniqueness theorem. $\square$

Remark 13.14 (What deficient rank does and does not prove)*.* If $D\mathcal{F}$ has constant rank $r < d$ nearby, the constant-rank theorem gives a local target fiber of dimension $d-r$, tangent to $\ker D\mathcal{F}$. Without constant rank or explicit target-preserving curves, kernel vectors are only infinitesimal target-null directions; deficient rank alone does not rule out a singular isolated solution. The moduli-locking criterion in this paper nevertheless requires the stronger row-minor certificate so that isolation is public and stable under first-order perturbations.

Theorem 13.15 (Interval contraction existence and isolation receipt)**.** *Let $H : B \to \mathbb{R}^N$ be a square selected constraint-and-readout system on a closed rational box $B \subset \mathbb{R}^N$. Let $x_0 \in B$, let A be an invertible rational matrix, and let $[H(x_0)]$ and $[J(B)]$ be outward-rounded interval enclosures of the residual and every Jacobian on B. Define*

$$K(B) = x_0 - A[H(x_0)] + (I - A[J(B)])(B - x_0).$$

If

$$K(B) \subset \operatorname{int} B, \qquad \sup_{J \in [J(B)]} \|I - AJ\|_\infty \leq q < 1,$$

then H has exactly one zero in B. If the selected equations are proved to generate the full completion-and-target zero system on B, this is a full existence and isolation certificate. Otherwise every omitted required row needs an independent identity or exact full-system solution receipt at that root. An interval determinant excluding zero certifies rank only; it does not by itself certify existence.

Proof. Set $T(x) = x - AH(x)$. The interval mean-value enclosure and the definition of $K(B)$ give $T(B) \subset K(B) \subset B$. The norm bound makes T a contraction on the complete metric space B. Banach's fixed-point theorem gives exactly one fixed point. Since A is invertible, $T(x) = x$ is equivalent to $H(x) = 0$. The statements about omitted rows and a determinant follow because neither condition proves those rows vanish or proves that a zero exists. $\square$

Theorem 13.16 (Branch-global cover certificate)**.** *Let a declared physical branch domain be covered by finitely many validated quotient-chart boxes. Assume one box has the full interval existence-and-isolation receipt of Theorem 13.15. Assume every other box is excluded because at least one required residual interval omits zero, and assume chart boundaries, singular strata, all discrete completion choices, and every noncompact end are covered or excluded by certified tail bounds. Then the declared branch contains exactly one target-matching physical orbit. If any box, boundary, stratum, discrete choice, or end is unresolved, the result is local and branch-global uniqueness does not follow.*

Proof. The validated cover places every point of the declared branch in one of the listed regions. The accepted box contains exactly one physical orbit. Every other region contains no joint zero by its residual exclusion certificate. Boundary, singular, discrete, and tail hypotheses remove the places omitted by ordinary interior chart boxes. Hence no second orbit exists on the declared branch. An unresolved region invalidates the exhaustive disjunction and gives the final statement. $\square$

13.9 Problem 9: operator safety

Gauge invariance allows the standard dangerous operators

$$LH_u, \quad LLe^c, \quad LQd^c, \quad u^c d^c d^c, \quad QQQL, \quad u^c u^c d^c e^c.$$

The operator-safe proposal adds the MSSM $\mathbb{Z}_4^R$ safety layer. Lee, Raby, Ratz, Ross, Schieren, Schmidt-Hoberg, and Vaudrevange prove that, allowing Green-Schwarz anomaly cancellation and requiring a discrete symmetry commuting with $SO(10)$ that forbids the perturbative μ-term, the MSSM has a unique $\mathbb{Z}_4^R$ symmetry. It leaves exact matter parity after nonperturbative breaking and suppresses dimension-five baryon/lepton violation [25].

Use the charge assignment

$$R(Q) = R(u^c) = R(d^c) = R(L) = R(e^c) = 1, \qquad R(H_u) = R(H_d) = 0, \qquad R(W) = 2 \pmod 4.$$

A perturbative superpotential monomial is allowed precisely when its total R-charge is 2 (mod 4).

Operator	$\mathbb{Z}_4^R$ **charge**	**Result**
$QH_u u^c$	$1+0+1=2$	Up-type Yukawa allowed.
$QH_d d^c$	$1+0+1=2$	Down-type Yukawa channel allowed; coefficient not fixed.
$LH_d e^c$	$1+0+1=2$	Charged-lepton Yukawa allowed.
$LH_u LH_u$	$1+0+1+0=2$	Weinberg neutrino operator allowed.
LH_u	$1+0=1$	Bilinear RPV forbidden perturbatively.

LLe^c	$1+1+1=3$	Lepton-number RPV forbidden perturbatively.
LQd^c	$1+1+1=3$	Lepton-number RPV forbidden perturbatively.
$u^cd^cd^c$	$1+1+1=3$	Baryon-number RPV forbidden perturbatively.
$QQQL$	$1+1+1+1=0$	Dimension-five proton decay forbidden perturbatively.
$u^cu^cd^ce^c$	$1+1+1+1=0$	Dimension-five proton decay forbidden perturbatively.
H_uH_d	$0+0=0$	Perturbative μ-term forbidden.

Theorem 13.17 (OPH operator-safety theorem)**.** *At the MSSM charge-algebra level, adding the $\mathbb{Z}_4^R$ layer to $BD^{\mathrm{OPH}}_{n=1}$ permits all Standard Model Yukawa couplings and the Weinberg operator, forbids perturbative dimension-four RPV, forbids perturbative dimension-five proton decay, forbids the perturbative μ-term, and leaves exact matter parity after nonperturbative breaking. The charge-algebra proposal is*

$$BD^{\mathrm{OPH}}_{n=1,+} = BD^{\mathrm{OPH}}_{n=1} + \mathbb{Z}_4^R.$$

Proof. The charge table verifies the perturbative selection rule term by term. The allowed Yukawa and Weinberg operators have total R-charge 2 (mod 4). The RPV operators, the dimension-five proton-decay operators, and H_uH_d have total charge 1, 3, or 0 (mod 4), hence fail the 2 (mod 4) superpotential rule. The cited MSSM theorem supplies uniqueness of the $\mathbb{Z}_4^R$ safety layer under the stated anomaly and $SO(10)$-commuting assumptions, plus the matter-parity remnant after nonperturbative breaking. □

14 Necessary Conditions for the Named Proposal

The named operator-safe proposal must satisfy the same conditions as any competing string representative. The moduli calculation shows that it does not:

Condition	**Requirement**
Global group	The realized group is $(\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1))/\mathbb{Z}_6$.
Hypercharge	The charge lattice is the Standard Model lattice.
Color and generations	$N_c=3$ and $N_g=3$.
Higgs sector	The low-energy branch has one Higgs pair and decouples to the observed Higgs doublet.
Exotics	No light chiral exotics appear.
Extra visible gauge factors	No extra visible low-scale U(1) appears.
Connected unified-gauge adjoint	The product-group adjoint contains no mixed $(3,2,\pm 5/6)$ generator.
Operator safety	The MSSM $\mathbb{Z}_4^R$ safety layer forbids perturbative RPV, dimension-five proton decay, and the perturbative μ-term, with exact matter parity after nonperturbative breaking.

Vacuum, moduli, and thresholds	The branch must supply a stable physical vacuum and derived threshold map, solve $\mathcal{H}_{BD,n=1,+}(m_\star) = 0$, and certify an invertible augmented row minor plus an interval existence-and-isolation box. The fixed five-row proxy fails this condition on the documented 142-real-dimensional pre-completion slice; the completion constraints, stable point, and physical map are not supplied.

Theorem 14.1 (No mixed X/Y connected-adjoint generator)**.** *On the product-group branch, the connected gauge adjoint is*

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0).$$

The adjoint contains no mixed $(3, 2, \pm 5/6)$ *generator.*

This theorem is an algebraic statement about the connected OPH product-group adjoint and does not by itself produce a connection action or a particle. The Bouchard–Donagi computation is a separate zero-mode calculation. It does not enumerate massive Wilson-line or Kaluza–Klein X/Y-type modes, calculate their couplings, or bound the proton decay operators obtained by integrating them out. Those questions belong to the heavy-spectrum and threshold certificate.

The executable calculation verifies the algebraic statements using exact rational arithmetic:

$$\mathrm{SU}(3)^2\mathrm{U}(1) = 0, \quad \mathrm{SU}(2)^2\mathrm{U}(1) = 0, \quad \mathrm{grav}^2\mathrm{U}(1) = 0, \quad \mathrm{U}(1)^3 = 0.$$

It also verifies the one-Higgs Yukawa hypercharge sums and records the $\mathbb{Z}_2$-cover generation arithmetic

$$|c_3(\widetilde{V})| = 12, \qquad |\Gamma| = 2, \qquad N_g = \frac{|c_3|}{2|\Gamma|} = 3.$$

Remark 14.2 (Operator safety certificate)*.* Gauge invariance leaves the usual MSSM danger operators hypercharge-neutral:

$$LH_u, \quad LLe^c, \quad LQd^c, \quad u^cd^cd^c, \quad QQQL, \quad u^cu^cd^ce^c.$$

Bouchard, Cvetic, and Donagi computed the classical trilinear couplings for this heterotic MSSM and report nonzero up-sector Yukawa couplings, vanishing R-parity-violating terms, and proton stability at that trilinear level [21]. The $\mathbb{Z}_4^R$ layer proves the MSSM charge-algebra safety statement. A BD certificate must realize this safety layer, or an equivalent selection rule, directly in the compactification data.

15 Threshold Proxy and Moduli-Locking Target

The published Bouchard–Donagi calculation fixes a visible zero-mode spectrum. It does not supply the physical vacuum point or solve the nonzero-mode problem. The source inventory contains eleven Kähler moduli, eleven complex-structure moduli, and fifty-one invariant bundle moduli, but no stabilized values or mass matrices [21, 22]. The original strongly coupled completion has a trivial hidden bundle and unbroken hidden E_8; bulk M5-branes cancel the remaining anomaly class. Perturbative hidden-bundle variants exist, so the hidden completion is itself a branch choice [20, 22]. None of these papers supplies a branch-derived string scale, full heavy spectrum, threshold determinant, or low-energy spectrum run.

The OPH low-energy target used here has no supersymmetric partner sector. The interface problem arises because the BD compactification is $N = 1$ supersymmetric. A complete certificate must therefore supply a decoupling map. A conventional route needs supersymmetry breaking, mediation, and soft boundary conditions. A non-supersymmetric route would instead require a separate, independently consistent UV construction or deformation. For this BD row to pass, the continuation must be proved OPH-equivalent to the cited BD visible-branch data, as well as establishing worldsheet or modular consistency, anomaly cancellation, a stable vacuum, its spectrum and couplings, and the resulting thresholds. An unrelated construction is a different candidate, and a projection label or hash is not sufficient.

The quantities below are target-side one-loop proxies. They do not use a BD heavy spectrum, stabilized moduli point, decoupling map, hidden-sector dynamics, mediation data, soft terms, or spectrum-generator output. Proxy reproduction is therefore not threshold compatibility, and $BD^{\mathrm{OPH}}_{n=1,+}$ is only a named structural row; it fails the selection condition. The accompanying machine-readable threshold-spectrum bundle records its dependency provenance, cryptographic hashes, selectors, an 80-decimal-digit arithmetic policy, scheme declaration, and the negative selection result.

From reference-side coordinates, including candidate-only Higgs/top inputs, the proxy computes

Quantity	**Proxy coordinate from declared inputs**
Tree-level top coordinate	$y_t^{\mathrm{tree}} = 0.987745211164$
Tree-level Higgs quartic coordinate	$\lambda_H^{\mathrm{tree}} = 0.128706603202$
MSSM tree-level quartic ceiling	$\lambda_{\mathrm{MSSM,max}} = 0.068973725409$
Large-tan β algebraic quartic gap	$\Delta\lambda_{\mathrm{proxy}} = 0.059732877792$
Tree-level Higgs mass proxy ceiling	$m_{h,\mathrm{tree}}^{\mathrm{max}} = 91.652460286\,\mathrm{GeV}$

Equivalently, the tree-level coordinates are

$$y_t^{\mathrm{tree}} = 0.987745211164, \qquad \lambda_H^{\mathrm{tree}} = 0.128706603202,$$

with the MSSM tree-level quartic ceiling

$$\lambda_{\mathrm{MSSM,max}} = \frac{\pi}{2}(\alpha_2(m_Z) + \alpha_Y(m_Z)) = 0.068973725409,$$

and the corresponding algebraic gap is

$$\Delta\lambda_{\mathrm{proxy}} = 0.059732877792.$$

The displayed y_t^{tree} is likewise the algebraic pole-coordinate conversion $\sqrt{2}m_t/v$, not a running Yukawa in a common scheme. The quartic subtraction mixes declared surface or pole coordinates with a tree-level MSSM expression and does not define a running quartic in a common renormalization scheme.

15.1 One-loop stop proxy for the conventional breaking route

Using

$$\Delta m_h^2 \simeq \frac{3m_t^4}{2\pi^2 v^2}\left[\log\frac{M_S^2}{m_t^2} + \frac{X_t^2}{M_S^2}\left(1 - \frac{X_t^2}{12M_S^2}\right)\right],$$

and the tree-level electroweak ceiling above, the fixed proxy gives:

$\tan\beta$	X_t/M_S	M_S proxy target
2	0	3046.266 GeV
2	$\sqrt{6}$	679.714 GeV
5	0	1191.830 GeV
5	$\sqrt{6}$	265.933 GeV
10	0	968.658 GeV
10	$\sqrt{6}$	216.137 GeV
50	0	901.608 GeV
50	$\sqrt{6}$	201.176 GeV

These rows invert a leading one-loop formula. They do not determine $\tan\beta$, M_S, or X_t from the BD branch, and some rows have too little scale separation for a controlled leading-log interpretation. If the conventional breaking route is chosen, a full soft-term and spectrum calculation must test them in one running scheme. A non-supersymmetric alternative must derive its own masses and decoupling thresholds within the independently consistent construction.

15.2 Gauge-unification proxy

A one-loop SM/MSSM running proxy with $M_{\mathrm{SUSY}} = 1\,\mathrm{TeV}$ and $\alpha_1 = (5/3)\alpha_Y$ gives

$$\log_{10}(M_U/\mathrm{GeV}) = 16.0815807506,$$

$$M_U = 1.20665 \times 10^{16}\,\mathrm{GeV}, \qquad \alpha_U^{-1} = 26.0176813615,$$

$$\alpha_3^{\mathrm{pred}}(m_Z) = 0.111511308019.$$

The symbols α_U and M_U here name the one-loop running-unification proxy coordinates of this subsection; they are different objects from the finite-screen unified gauge width $\alpha_U(P)$ of the fine-structure paper and the capacity-bridge α_U of The Standard Model gauge paper, which carry the same symbols with different values. Against the PDG 2025 comparison value $\alpha_s(m_Z) = 0.1180 \pm 0.0009$ [24], the residual inverse-coupling combination is

$$\mathcal{C}_3^{\mathrm{proxy}} = -0.493124115142,$$

with the reference-only interval

$$-0.557271454453 \le \mathcal{C}_3^{\mathrm{proxy}} \le -0.427990736457.$$

Here the declared matching convention is

$$\mathcal{C}_3^{\mathrm{proxy}} = \alpha_s(m_Z)^{-1} - \left[\alpha_3^{\mathrm{pred}}(m_Z)\right]^{-1}$$

after the one-loop SM/MSSM step running. This is not a computed BD string/GUT threshold. Scanning the arbitrary common M_{SUSY} step from 0.5 to 10 TeV moves the central residual from about -0.3434 to -0.9905. A heavy spectrum and a fixed matching prescription are needed before a physical threshold corridor exists.

15.3 Finite structural comparison

The encoded structural comparison scores representative branches against the declared criteria. Threshold and moduli conditions are excluded from this finite structural score:

Candidate	Score	Classification
$BD_{n=0}^{\mathrm{SU}(5),\mathbb{Z}_2}$	7/9	Rejected: no Higgs pair.
$BD_{n=1}^{\mathrm{SU}(5),\mathbb{Z}_2}$	8/9	Geometric witness; safety layer absent.
$BD_{n=1,+}^{\mathrm{SU}(5),\mathbb{Z}_2}$	9/9	Unique full-score structural row; not an OPH-correct witness.
$BD_{n=2}^{\mathrm{SU}(5),\mathbb{Z}_2}$	7/9	Fails the zero-mode minimality score.
BHOP $\mathrm{SU}(4), \mathbb{Z}_3 \times \mathbb{Z}_3$	4/9	Backup witness.
Generic $\mathrm{Spin}(32)/\mathbb{Z}_2$ heterotic	0/9	Rejected as minimal class.

Inside this encoded structural comparison, $BD_{n=1}^{\mathrm{SU}(5),\mathbb{Z}_2}$ is the geometric zero-mode row and $BD_{n=1,+}^{\mathrm{SU}(5),\mathbb{Z}_2}$ is the unique structural full-score row. The score excludes the threshold and moduli conditions and therefore cannot select an OPH-correct string witness. Corollary 3.5 gives the nonselection result.

Theorem 15.1 (Encoded structural comparison uniqueness)**.** *Let $\mathcal{C}_{\mathrm{cmp}}$ be the six-branch candidate family in the table above, scored against the nine encoded structural criteria. The unique full-score row in $\mathcal{C}_{\mathrm{cmp}}$ is $BD_{n=1,+}^{\mathrm{SU}(5),\mathbb{Z}_2}$. No threshold, moduli, or low-energy passage follows from this finite score.*

Proof. The table assigns scores $7/9, 8/9, 9/9, 7/9, 4/9, 0/9$ to the six listed candidates. The only score equal to the structural criterion count is the $BD_{n=1,+}^{\mathrm{SU}(5),\mathbb{Z}_2}$ row. The full-score structural set inside $\mathcal{C}_{\mathrm{cmp}}$ is the singleton

$$\{BD_{n=1,+}^{\mathrm{SU}(5),\mathbb{Z}_2}\}.$$

□

These proxy numbers define a target-side screen for a full cohomology, Yukawa, heavy-spectrum, threshold, and decoupling computation. The conventional MSSM route requires supersymmetry breaking and soft boundary conditions. A non-supersymmetric route could rehabilitate this BD row only if it is an independently consistent, OPH-equivalent deformation or continuation that preserves the cited BD visible-branch data; an unrelated construction is a different candidate. Cohomology algorithms such as cohomCalg-style line-bundle cohomology are designed for massless-mode calculations in compactifications [30]. Spectrum tools such as SOFTSUSY solve MSSM renormalization-group equations with supplied conventional breaking constraints [31]; citing such a tool is not evidence that a BD spectrum run occurred.

16 Physical Moduli Slice and Rank/Isolation Certificate

The moduli criterion must begin with a source space. Quotienting by the kernel of a short observable list would make full rank tautological, so the quotient and the target map are defined separately.

Definition 16.1 (Published and completed physical BD slices)**.** *Fix the BD topological and equivariant branch and its one-massless-Higgs-pair cohomology stratum. Let* $\widetilde{\mathcal{M}}_{\rm pub}^{n=1}$ *be the smooth published visible-sector solution deformations satisfying Hermitian Yang–Mills, bundle stability, the fixed topological and equivariant data, and the one-Higgs condition. Quotient only independently verified OPH-invisible transformations: diffeomorphism and bundle/B-field gauge orbits, bundle isomorphisms, and changes of local representative or exact field basis. The documented pre-completion physical slice is*

$$\mathcal{M}_{\rm pub}^{n=1} = \widetilde{\mathcal{M}}_{\rm pub}^{n=1}/\mathcal{G}_{\rm pub,inv}.$$

For an operator-safe completed branch, let $q = (s, h, w, \theta)$ *collect the dilaton, hidden-sector, bulk-five-brane, and scheme-locked threshold/decoupling data. Given a realized safety layer and a fixed completion route, define*

$$\mathcal{M}^{\rm phys}_{BD,n=1,+} = \left\{([m], q) \in \mathcal{M}_{\rm pub}^{n=1} \times \mathcal{D}_{\rm comp} : \mathcal{C}_{\rm comp}([m], q) = 0\right\} / \mathcal{G}_{\rm comp,inv},$$

where $\mathcal{C}_{\rm comp} = 0$ *contains the hidden/anomaly, safety-realization, spectrum, threshold, vacuum, stabilization, and decoupling equations. The quotient* $\mathcal{G}_{\rm comp,inv}$ *contains only exact scheme or duality-presentation changes with identical complete OPH readout. Genuine mediation or boundary parameters are physical source coordinates. Heavy masses and thresholds are derived on this constraint locus; benchmark values are not independent fit coordinates.*

At a smooth point p*, a physical slice* S_p *is a local representative of this quotient, and*

$$T_{[p]}\mathcal{M}^{\rm phys}_{BD,n=1,+} \cong \frac{\ker D\mathcal{C}_{\rm comp}(p)}{T_p(\mathcal{G}_{\rm comp,inv} \cdot p)}.$$

Its real dimension d_p *is part of the completion certificate. It cannot be inferred from the ambient published-moduli count because stabilization and vacuum equations can remove tangent directions. A physical modulus is not put in either invisible quotient merely because the fixed target packet omits its observable effects.*

Definition 16.2 (Constraint-augmented BD selector map)**.** *Let* $\mathcal{X}_{BD,n=1,+}$ *be a local quotient chart for* $\mathcal{M}_{\rm pub}^{n=1} \times \mathcal{D}_{\rm comp}$ *after removing only the certified presentation redundancies, but before imposing* $\mathcal{C}_{\rm comp} = 0$*. Once the physical forward calculation exists, define*

$$\mathcal{H}_{BD,n=1,+} = (\mathcal{C}_{\rm comp}, \mathcal{F}_{BD,n=1,+} - \mathcal{O}_{\rm OPH}).$$

The completion rows include stationarity and all branch-consistency equations. The target rows are observer-visible quantities fixed independently of the candidate. Discrete topology, criticality, cohomology, and operator-safety conditions are separate Boolean certificates. A positive local receipt may either apply Theorem 13.13 on the completed physical slice or apply Theorem 13.10 directly to this ambient quotient chart. Theorem 13.15 supplies the stronger existence-and-uniqueness form.

Proposition 16.3 (Ambient one-Higgs normal dimension and conditional pullback rank)**.** *For the invariant BD extension data, the Higgs cohomology is controlled by the negative block* $M_- : \mathbb{C}^8 \to \mathbb{C}^9$*, with*

$$n = 1 \quad \Longleftrightarrow \quad \operatorname{rank} M_- = 7$$

while the positive 9×9 *block is full rank. At a rank-seven matrix, the ambient determinantal normal space is*

$$\operatorname{Hom}(\ker M_-, \operatorname{coker} M_-), \qquad \dim_{\mathbb{C}} = 1 \cdot 2 = 2.$$

If the cited BD pullback is scheme-theoretically regular and transverse at a chosen point, its local defining map has exact transverse complex rank two. The published codimension count and smooth reduced-locus description do not by themselves prove this differential statement. A positive pullback-rank receipt must supply local extension coordinates, the entries of M_-, two local defining equations, and a nonzero exact or interval-certified 2×2 Jacobian minor. Under that receipt, the two normal extension directions are OPH-visible branch-exit directions: leaving the locus changes the one-massless-Higgs-pair data.

Proof. At rank seven, $\dim_{\mathbb{C}} \ker M_- = 8 - 7 = 1$ and $\dim_{\mathbb{C}} \operatorname{coker} M_- = 9 - 7 = 2$. The standard tangent space to a fixed-rank matrix stratum consists of perturbations δM for which the induced map $\ker M_- \to \operatorname{coker} M_-$ vanishes. The quotient by that tangent space is therefore the displayed two-complex-dimensional Hom space. Ambient rank alone does not prove transversality of the extension-moduli pullback. Equal codimension also does not exclude a nonreduced pullback whose derivative loses rank. Surjectivity of the induced normal map is exactly the additional hypothesis that yields transverse complex rank two. The cited calculations give the one-Higgs codimension and identify the two relevant normal modulus couplings [20, 21]; the machine packet does not contain the local map entries needed to verify the additional hypothesis. □

The exact invariant bundle-moduli count is 51 complex dimensions. Restricting to the smooth codimension-two one-Higgs locus leaves $49 = 42 + 4 + 3$ complex tangent directions: forty-two extension tangents and seven constituent-bundle deformations. Together with eleven complexified Kähler and eleven complex-structure moduli, the documented pre-completion source slice is

$$\dim_{\mathbb{C}} \mathcal{M}^{n=1}_{\mathrm{pub}} = 11 + 11 + (51 - 2) = 71, \qquad \dim_{\mathbb{R}} \mathcal{M}^{n=1}_{\mathrm{pub}} = 142.$$

The $\mathbb{Z}_4^R$ label is discrete and adds no tangent direction. At a singular point of the determinantal locus the Zariski tangent dimension can be larger. The displayed 142 is not a lower bound for $\dim_{\mathbb{R}} \mathcal{M}^{\mathrm{phys}}_{BD,n=1,+}$: completion variables can add ambient coordinates, while independent stabilization and vacuum equations can cut them.

Definition 16.4 (Coordinate-locked BD comparison map)**.** *Separate the discrete structural target from the differentiable target. The global group, generation count, hypercharge lattice, Higgs count, exotic-matter exclusion, and safety rule define the allowed stratum; locally constant discrete conditions do not supply rows of a Jacobian. On a physical slice and a completed threshold receipt define*

$$\begin{aligned} \mathcal{F}_{BD,n=1} \equiv \mathcal{F}_{BD,n=1,+} : \mathcal{M}^{\mathrm{phys}}_{BD,n=1,+} \longrightarrow \mathbb{R}^5, \\ [m, \theta] \longmapsto \left(\alpha_2^{BD}(m_Z), \alpha_Y^{BD}(m_Z), \frac{v^{BD}}{\mathrm{GeV}}, \frac{m_H^{BD,\mathrm{pole}}}{\mathrm{GeV}}, \frac{m_t^{BD,\mathrm{pole}}}{\mathrm{GeV}} \right), \end{aligned}$$

with α_2 and α_Y evaluated in their declared running scheme and at their declared scale, v in its declared normalization, and m_H, m_t as declared pole coordinates. The packet does not fix a common threshold scheme, so this is a comparison contract rather than an executable physical map. Its ordered comparison vector is

$$\mathcal{O}^{(5)}_{\mathrm{OPH,cmp}} = \begin{pmatrix} 0.03377843630219015 \\ 0.010131601067241624 \\ 246.76711732749683 \\ 125.1995304097179 \\ 172.3523553288312 \end{pmatrix}.$$

The first three entries are declared OPH surface coordinates. The last two are explicitly candidate-only rather than OPH predictions, so the displayed vector is not a complete OPH target. The compare-only $\alpha_3(m_Z)$ and m_Z inputs are not additional target rows. A physical $\mathcal{F}_{BD,n=1,+}$ exists only after the same branch supplies the hidden/nonzero-mode spectrum, normalized Yukawas, matching scales and scheme, and decoupling map. Genuine mediation or boundary parameters are physical source coordinates; convention-only scheme coordinates are quotiented.

Definition 16.5 (Admissible target-rank row)**.** *A row may contribute to the target-rank certificate only when its observable definition, units, scheme, scale, acceptance region, and claim scope are fixed independently of the candidate evaluation. The row must be observer-visible and source-derived. A candidate benchmark, fitted nuisance coordinate, or target value copied from the evaluated point is not an OPH target row. Physical mediation and boundary parameters are source coordinates. The evaluator freezes and hashes the source model before it loads the comparison data.*

Lemma 16.6 (No self-targeting rank certificate)**.** *Appending candidate coordinates to a readout and assigning their values at a chosen point can manufacture full rank at any regular point. Such rows provide no selection evidence. In particular, the candidate-only Higgs and top coordinates in the five-row comparison packet do not count toward an OPH target-rank certificate. They can be used as independently fixed empirical acceptance tests, with declared covariance and scheme. They support an OPH prediction only if a separate source theorem derives them.*

Proof. In local physical coordinates $x_1, \ldots, x_d$, append the map $x \mapsto (x_1, \ldots, x_d)$ and choose target values $x_i(p)$. Its derivative is the identity, so the augmented readout has a nonsingular $d \times d$ minor regardless of the physical theory. Independent precommitment excludes this construction. The fixed packet classifies its Higgs and top values as candidate-only, which gives the stated application. □

Theorem 16.7 (Published-slice transverse-rank obstruction)**.** *For the source and target in Definition 16.4, the published input data do not supply the completed constraint locus, its dimension $d_\star$, a point $m_\star$, an evaluable physical $\mathcal{F}_{BD,n=1,+}$ or $\mathcal{H}_{BD,n=1,+}$, or either Jacobian. The only executable object is a target-side proxy that uses no BD branch value. Pulled back to the published BD slice, that proxy is constant, and hence*

$$D\mathcal{F}_{\text{proxy}} = 0_{5\times 142}, \qquad \operatorname{rank} D\mathcal{F}_{\text{proxy}} = 0.$$

Its proxy target fiber contains the entire local published slice and is therefore not isolated. Any differentiable map defined on all of $\mathcal{M}^{n=1}_{\text{pub}}$ with this same five-real-coordinate codomain satisfies

$$\operatorname{rank} D\mathcal{F} \leq 5 < 142, \qquad \dim\ker D\mathcal{F} \geq 137.$$

These equations are a rank obstruction on the published pre-completion slice, not a dimension claim about an unprovided stabilized slice. A completion could cut the physical dimension to five or fewer, but it must certify those constraints and the resulting Jacobian. No such certificate is present, so $BD^{\text{OPH}}_{n=1,+}$ fails this paper's moduli-locking condition. The two-complex-dimensional ambient normal calculation in Proposition 16.3 concerns only the Higgs multiplicity locus, and its pullback transversality is not established. It does not change this conclusion.

Proof. The published data record 51 invariant bundle, 11 complex-structure, and 11 complexified Kähler moduli, with a codimension-two one-Higgs locus. This gives the stated 142-real-dimensional published slice. The packet records no completed slice, selected point, or physical Jacobian. The proxy receipt states that no BD branch values enter, so its pullback factors through a point and has the displayed zero derivative. For any map from the full published slice to $\mathbb{R}^5$, elementary linear algebra gives rank at most five and rank–nullity gives nullity at least $142 - 5 = 137$. Neither calculation supplies the absent completed-slice row minor required by Theorem 13.13 or the augmented constraint certificate of Theorem 13.10. □

The 137-dimensional bound concerns maps on the full published pre-completion slice. It is not a nullity bound after unknown completion constraints. For a nonconstant map, kernel vectors become proved adjustable flat curves only after a constant-rank neighborhood or explicit target-preserving family is supplied. Every retained published direction is an actual flat direction of the stated constant proxy.

Direction family	**Classification**	**Physical meaning and scope**
Eleven complexified Kähler directions	OPH-visible published moduli	Volumes and B-field axions enter G_4, gauge kinetic functions, KK/winding scales, and thresholds. Proxy-flat; completion effects are not determined.
Eleven complex-structure directions	OPH-visible published moduli	Periods, spectra, normalized Yukawas, and thresholds. Proxy-flat; completion effects are not determined.
Forty-nine complex bundle tangents to $n = 1$	OPH-visible published moduli	Bundle cohomology, singlet couplings, Yukawas, operator rules, and heavy/vectorlike masses. Proxy-flat; completion effects are not determined.
Two complex ambient bundle normals	OPH-visible branch-exit candidates	The determinantal normal space has complex dimension two. Pullback transversality requires the local Jacobian receipt. Under that receipt, moving normally changes the Higgs/lepton mass pairing.
Dilaton, hidden/M5, and decoupling sectors	OPH-visible completion data, uncounted	Their variables and constraint equations are missing, so their net effect on the completed-slice dimension is unknown.
Gauge, diffeomorphism, bundle-isomorphism, exact basis/scheme, and proved duality directions	OPH-invisible	Presentation redundancies are quotiented; they do not rescue the rank count.

Corollary 16.8 (Scope of the rank certificate)**.** *The transverse-rank certificate rejects $BD^{\mathrm{OPH}}_{n=1,+}$ because the stated proxy has rank zero on the published slice and no completed-slice rank or isolation receipt exists. This certificate does not exclude a completion whose proved constraints leave a smaller physical slice; such a completion is absent from the stated candidate data. The cited BD one-Higgs zero-mode construction is a structural benchmark. The independently stated OPH recovered core has different premises. A positive certificate requires the completion equations, their stability domain, a selected physical point, a derived threshold/decoupling map with precommitted rows, and either the completed-slice certificate of Theorem 13.13 or the constraint-augmented certificate of Theorem 13.10. Existence and uniqueness in a numerical box require Theorem 13.15; a floating-point matrix-rank call alone is not a proof.*

17 Conditional BD Witness Certificate Criterion

The strongest mathematically clean physical statement has certificate form. It separates what the paper proves by algebra from what a complete BD branch must certify. Section 16 proves the negative rank result. The stated data do not satisfy the positive witness premises. No global uniqueness statement follows without a complete comparison catalogue.

Theorem 17.1 (Conditional OPH BD witness implication)**.** *Assume the following data.*

1. *The declared OPH selector target is*

$$\mathfrak{S}_{\mathrm{OPH}} = \begin{pmatrix} (\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1))/\mathbb{Z}_6, \quad N_c = 3, \quad N_g = 3, \quad Y_{\mathrm{SM}}, \\ n_H = 1, \quad \textit{no light chiral exotics}, \quad \mathbb{Z}_4^R \textit{ safety} \end{pmatrix}.$$

   Within this target, $N_c = 3$, hypercharge, and the maximal faithful image are consequences of the conditional matrix current and matter theorem. Physical selection of the $\mathbb{Z}_6$ quotient is not claimed. $N_g = 3$, one-Higgs, no-light-exotics, and safety are independent selector assumptions.

2. *The effective string continuation is read through the heterotic edge-sector branch and satisfies the finite-carrier critical-edge certificate of Section 9.*

3. *The BD cohomology certificate realizes the massless one-Higgs, three-generation, no-visible-exotics stratum with Wilson-line centralizer $S(\mathrm{U}(3)\times\mathrm{U}(2))$.*

4. *The same branch realizes $\mathbb{Z}_4^R$, or an OPH-equivalent safety rule, on the visible operator algebra.*

5. *A threshold/spectrum receipt supplies the nonzero-mode and hidden spectra, the dilaton and other stabilized moduli, the bulk five-brane sector or a proof of its absence, normalized Yukawa boundary data, physical scales, and a map that decouples the supersymmetric compactification into the non-supersymmetric low-energy branch. For a conventional breaking route this includes mediation and soft boundary conditions. A non-supersymmetric alternative must instead supply an independently consistent, OPH-equivalent deformation or continuation of the BD branch, prove that it preserves the cited visible cohomology and safety data, and compute its threshold map. An unrelated construction does not certify the BD row.*

6. *There is a certified quotient chart $\mathcal{X}_{BD,n=1,+}$, a complete constraint map $\mathcal{C}_{\mathrm{comp}}$, a precommitted scheme-locked target map, and a point $m_\star$ such that*

$$\mathcal{C}_{\mathrm{comp}}(m_\star) = 0, \qquad \mathcal{F}_{BD,n=1,+}(m_\star) = \mathcal{O}_{\mathrm{OPH}}.$$

   The same receipt proves dynamical stability under Theorem 13.12 and gives either an exact nonsingular square minor of the augmented map in Definition 16.2, or an interval contraction box satisfying Theorem 13.15. Every row obeys the no-self-targeting rule, and every omitted required equation follows by a certified identity or independent solution receipt.

Then the following OPH-stable equivalence class is a passing named critical-string witness:

$$\boxed{\left[BD^{\mathrm{SU}(5),\mathbb{Z}_2}_{n=1} + \mathbb{Z}_4^R\right]_{\mathrm{OPH}} = BD^{\mathrm{OPH}}_{n=1,+}.}$$

Proof. The first hypothesis declares the complete selector target. The complete-response theorem fixes its local Standard Model gauge Lie algebra. The conditional matrix current and matter theorem fix the hypercharge lattice, color count, common kernel, and maximal faithful image. Source selection of the current, matter action, and physical $\mathbb{Z}_6$ quotient is not claimed. Laboratory and continuum attachment of that quotient, generation count, Higgs count, exotic-matter condition, and operator-safety condition are explicit selector assumptions. The heterotic edge-sector branch and the finite-carrier critical-edge certificate place the named witness on the critical edge route. The BD certificate supplies a nonempty $\mathbb{Z}_2$-quotient SU(5)-bundle witness with the required three-generation, no-exotics, one-Higgs massless cohomology. The centralizer theorem identifies its Wilson-line centralizer with the conditionally declared global form in the selector target. The $\mathbb{Z}_4^R$ safety layer establishes the visible operator algebra condition. The threshold/spectrum and decoupling hypothesis supplies the physical map from that supersymmetric zero-mode branch to the OPH target. The stability receipt gives a physical vacuum, while the augmented exact or interval certificate isolates $m_\star$ modulo independently verified OPH-invisible equivalence. Therefore the named class

$$\left[BD_{n=1}^{\mathrm{SU}(5),\mathbb{Z}_2} + \mathbb{Z}_4^R\right]_{\mathrm{OPH}}$$

satisfies the OPH-correct witness conditions. □

Remark 17.2 (Certificate boundary). The theorem gives the witness proof shape. Its algebraic steps close in this paper. The critical-edge receipt table, cohomology table, $\mathbb{Z}_4^R$-realization table, Yukawa/superpotential table, heavy-spectrum and decoupling table, threshold table, stability table, and augmented interval-isolation table would form the premises of a positive witness theorem. The available data instead establish a rank obstruction and its failure scope. A comparison table over an exhaustive candidate class is an additional requirement after an existence row passes.

Theorem 17.3 (Certificate closure criterion)**.** *A named local BD witness passes only when critical-edge closure, cohomology, $\mathbb{Z}_4^R$ realization, superpotential safety, threshold and low-energy matching, completed-slice stability, and augmented local isolation all pass. Branch-global uniqueness additionally requires a validated cover of every chart, boundary, singular stratum, discrete completion, and noncompact end of that BD branch. The class-restricted equality*

$$\{\mathcal{T} \in \mathcal{C} : \mathcal{T} \text{ is OPH-correct}\}/\sim_{\mathrm{OPH}} = \{BD_{n=1,+}^{\mathrm{OPH}}\}$$

additionally requires exhaustive catalogue coverage, duality deduplication, one replayed verdict for every equivalence class, exactly one passing class, and no inconclusive class. A claim beyond $\mathcal{C}$ requires a separate theorem that the declared candidate universe is covered by $\mathcal{C}/\sim_{\mathrm{OPH}}$.

Proof. The first conjunction is exactly the local witness definition and the local existence-and-isolation theorems. A validated branch cover excludes any second solution on the same declared branch. The exhaustive catalogue then gives the exact set of passing equivalence classes because every row is covered and none is inconclusive. Singleton membership gives the class-restricted equality. The last conclusion follows only when the coverage theorem maps every candidate in the wider declared universe into that catalogue. □

Remark 17.4 (Closure boundary). The transverse-rank certificate fails the moduli-locking condition for the five-coordinate contract. Thus the closure criterion is not satisfied by the BD row in the published input data; comparative uniqueness cannot select a row that fails an existence condition.

18 Empirical Test Surface

The BD structural row is useful because its structural claims and assigned proxy targets can fail. The table keeps those two classes separate. Published zero-mode results test the visible massless branch. Numerical rows define screens for the required heavy-spectrum and low-energy calculation, which is absent from the published data; they are not realized BD predictions.

Structural claim or proxy screen	**Numerical or structural target**	**Failure meaning**
Global Standard Model group	$(\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1))/\mathbb{Z}_6$.	Retracts the named visible branch.
Color and generations	$N_c = 3$, $N_g = 3$, with $\lvert c_3(\widetilde{V})\rvert = 12$, $\lvert\Gamma\rvert = 2$.	Retracts the BD witness.
Higgs sector	One massless Higgs pair in the published cohomology; its decoupling to the observed light Higgs requires a separate map.	Extra massless pairs retract the selected zero-mode stratum; a failed decoupling map blocks low-energy passage.
Visible exotics	No massless visible chiral exotics in the published cohomology.	Retracts the minimal zero-mode branch.
Visible gauge factors	No extra visible low-scale U(1).	Severe pressure on the normal-form sieve.
Connected unified-gauge adjoint	No mixed $(3, 2, \pm5/6)$ generator in the connected product-group adjoint.	A mixed generator conflicts with the product-group branch; connection dynamics and BD particle modes require separate checks.
Operator safety	$\mathbb{Z}_4^R$ permits Yukawas and the Weinberg operator, forbids perturbative RPV, dimension-five proton decay, and the perturbative μ-term, and leaves matter parity after nonperturbative breaking [25].	Safety certificate fails if the BD branch lacks this layer or an equivalent rule.
Classical RPV trilinears	BCD report vanishing R-parity-violating trilinears at the classical level in this heterotic MSSM [21]. Worldsheet-instanton corrections are outside that result.	Classical superpotential failure.
Tree-level top coordinate	$y_t^{\mathrm{tree}} = 0.987745211164$.	A completed forward spectrum misses the assigned proxy screen; no failure is inferred from the proxy alone.
Tree-level Higgs coordinate	$\lambda_H^{\mathrm{tree}} = 0.128706603202$.	A completed same-scheme calculation misses the assigned proxy screen.

Algebraic quartic gap	$\Delta\lambda_{\mathrm{proxy}} = 0.059732877792$.	A conventional breaking calculation fails this proxy screen; the row is not a heavy-spectrum certificate.
Stop proxy	The table gives M_S values for representative $(\tan\beta, X_t/M_S)$ choices.	Conventional-route soft terms miss the proxy screen. A BD-equivalent non-supersymmetric deformation has its own mass and threshold calculation.
One-loop gauge residual	At $M_{\mathrm{SUSY}} = 1\,\mathrm{TeV}$, $\mathcal{C}_3^{\mathrm{proxy}} = -0.493124115142$, with the reference-only interval shown in Section 15.	A supplied heavy-threshold combination misses the proxy screen; the residual itself is not a BD string threshold.

The published branch supplies a narrow massless visible package. This paper assigns numerical screens. The published data contain no threshold or spectrum certificate, so no threshold corridor is established for the BD row. The displayed five-coordinate contract is

$$\mathcal{F}_{BD,n=1,+}(m_\star) = \mathcal{O}^{(5)}_{\mathrm{OPH,cmp}},$$

but no $m_\star$ or physical forward map is supplied, and Theorem 16.7 proves that this five-row contract cannot meet the constraint-augmented locking condition on the documented slice without additional certified completion constraints.

19 Assumptions and Result Boundaries

The full string-model statement depends on the following explicit data:

Condition	**Required data**	**Result in this paper**
Central-charge and supercurrent certificate	Response-map rank $\operatorname{rank}\mathcal{R}_{\mathrm{port}} = 12$, positive $A_5 \times C_2$ covariance sectors, edge transfer OPEs, local graded half-repair $Q_N^2 \to H_R$, and torus/spin-structure receipt.	The non-implication theorem and finite critical-edge certificate theorem are proved. The carrier premises are not established.
Cohomology certificate	Explicit X, V, and equivariant $\mathbb{Z}_2$ action in cohomCalg, Sage, or Macaulay2 form.	The structural witness is cited; this paper does not reproduce the raw computation.
Safety-layer realization certificate	$\mathbb{Z}_4^R$, or an equivalent compactification selection rule, realized on the BD one-Higgs branch.	The MSSM charge algebra closes; BD realization is an additional premise.

Operator and superpotential certificate	Sheaf cup products, harmonic representatives, instanton data, and sector selection rules.	The BCD trilinear result supports RPV vanishing; no numerical texture follows from the stated data.
Threshold and spectrum certificate	Nonzero-mode and hidden spectra; stabilized Kähler, complex-structure, bundle, dilaton, and any bulk five-brane data; normalized Yukawa boundary data; string, compactification, and matching scales; and a low-energy decoupling map. A conventional breaking route also requires mediation and soft boundary conditions. A non-supersymmetric alternative requires an independently consistent UV construction or deformation and its own mass and threshold calculation.	The proxy coordinates reproduce, while compatibility is unevaluated. No heavy-threshold or low-energy spectrum certificate is present.
Vacuum-stability certificate	Complete effective potential, stationary point, physical scalar metric and Hessian, control bounds, and a positive mass or route-appropriate stability enclosure.	The stated data supply no BD stability result.
Moduli-locking certificate	Physical quotient, precommitted scheme-locked map, completed constraints, selected point, and a certified augmented Jacobian and interval-isolation box.	Certified negative result: the published pre-completion source has dimension 142 real, its five-row proxy has rank zero, and the completed source, point, and Jacobian are absent. The candidate fails selection.
Comparative uniqueness certificate	Branch-domain coverage, expanded heterotic edge-sector check, and any brane, orientifold, or duality-rewritten presentations claiming OPH equivalence.	The encoded comparison has one structural full-score row. It is not a selection or catalogue-coverage certificate.

The stated data contain no threshold or spectrum calculation, and the moduli-locking condition fails. Accordingly $BD^{\mathrm{OPH}}_{n=1,+}$ is only the full-score row of the declared structural comparison. It is neither the selected operator-safe string candidate nor a passing OPH-correct witness.

20 Falsifier Matrix

Failure mode	Consequence

Wrong global Standard Model group	Retracts the OPH visible landing branch.
Wrong hypercharge lattice	Retracts the named visible branch.
Fourth chiral generation	Retracts the realized branch.
Light chiral exotics	Retracts the BD one-Higgs witness.
Irreducible second light Higgs pair	Retracts the one-Higgs low-energy projection.
Extra visible low-scale U(1)	Severe pressure on the OPH normal-form sieve.
Mixed connected-adjoint X/Y generator	Conflicts with the connected product-group branch.
Massive BD X/Y-type modes or induced operators	Evaluated by the heavy-spectrum and coupling certificate; the zero-mode adjoint theorem does not decide them.
BD cohomology reproduction fails	Retracts $BD_{n=1}^{\mathrm{OPH}}$ as named geometric witness.
$\mathbb{Z}_4^R$ or equivalent safety layer absent	Retracts $BD_{n=1,+}^{\mathrm{OPH}}$ as an operator-safe proposal.
Dangerous operators survive without suppression	Severe pressure on the effective theory.
No moduli point matches $\mathcal{O}_{\mathrm{OPH}}$	Retracts the operator-safe selected candidate.
Constraint-augmented locking certificate fails or physical directions are unclassified	Retracts the operator-safe selected candidate; the structural BD row and recovered OPH core are separate claims and stand.
Central-charge or supercurrent condition fails	Retracts the heterotic critical-worldsheet identification of the sewn edge branch.
Critical-string lift fails	Weakens the string continuation; the OPH recovered-core theorems have different premises.

21 Logical Dependencies for De Sitter and String Theory

The argument has the following dependency order:

1. stable checkpoint continuation supplies the finite observer theorem;
2. finite-dimensional clock projection supplies the correlator statement;
3. T-holonomy is a $\mathbb{Z}_2^T$ cycle obstruction and a $w_1(L_T)$ class;
4. ordered checkpoint records select a local time orientation on the observer quotient;
5. edge-center record capacity gives $S_{\mathrm{dS}} = A/(4\ell_P^2) = N_{\mathrm{scr}}$;
6. the edge identity gives $Z_{\mathrm{edge}}(t) = K_t(1)$, with a controlled large-N_{edge} worldsheet branch;
7. the heterotic edge-polarization statement for the sewn-edge scaling limit requires OPEs, fermionic grading, spin structures, and no residual coset;
8. the vacuum sieve uses the global group identity, $\mathbb{Z}_4^R$ safety table, connected-adjoint theorem, and moduli-locking criterion;

9. a BD realization requires the $n = 1$ cohomology, $\mathbb{Z}_4^R$ realization, and operator catalogue as physical input;

10. a vacuum-selection conclusion additionally requires the nonzero-mode and hidden spectra, all physical moduli and scales, either a conventional SUSY-breaking decoupling route or an independently consistent OPH-equivalent non-supersymmetric deformation with its own mass and threshold calculation, the low-energy forward calculation, threshold matching, source hashes, and numerical precision. These data define the completed constraint map, physical Hessian, target map, augmented Jacobian, and interval-isolation box.

The observer and time-reversal statements use the first four dependencies. The critical-worldsheet and vacuum-selector statements additionally use the final four.

22 De Sitter and String-Theory Correspondence

The construction expresses several de Sitter and string-theory questions as a finite observer-visible normal-form problem for critical strings. Each conditional statement has an explicit premise.

Problem	**OPH statement**	**Required premise**
What is the physical observer in de Sitter?	A stable record-bearing patch subfederation with checkpoint continuation.	Record/collar model.
Why does a clock branch appear?	Clock time is a record-sector projection; imaginary semiclassical correlators appear after branch conditioning.	Observer-sector examples.
What is the clock-flip holonomy?	A $\mathbb{Z}_2^T$ cycle obstruction on the overlap nerve, $w_1(L_T)$.	Geometric examples.
Where is de Sitter entropy?	Edge-center record capacity, $S_{\mathrm{dS}} = A/(4\ell_P^2) = N_{\mathrm{scr}}$.	Capacity normalization.
Why do strings appear?	Sewn OPH edge partitions are heat-kernel sums; the controlled large-edge continuation is a worldsheet expansion.	Large-edge control.
Which string theory passes the OPH selector?	None under the stated premises. $BD^{\mathrm{OPH}}_{n=1,+}$ is the structural test row and fails selection: its proxy has rank zero and its completed physical slice and Jacobian are absent.	Critical-edge, cohomology, safety, threshold/spectrum, decoupling, and moduli certificates.
Why anthropic landscape selection is absent	OPH provides a target normal form $\Pi^{\mathrm{OPH}}_{\mathrm{IR}}(\mathcal{T}) = \mathfrak{S}_{\mathrm{OPH}}$, with no statistical sampling weight.	Exhaustive checks.

What fixes the SM global form?	The realized tensors have a common $\mathbb{Z}_6$ kernel and maximal faithful image $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$. They also descend through four central global forms. The six-axis menu matches the $\mathbb{Z}_6$ quotient after its coefficient relations are declared; the physical global form is not fixed by these results. A source-derived complete character category, same-source loop-to-kernel identity, and laboratory and continuum attachment are the premises of a fixed global form.	Conditional lattice receipt plus source, laboratory, and continuum attachment.
Why no connected simple-GUT X/Y generator?	The connected product-group adjoint has no $(3, 2, \pm 5/6)$ generator. Connection dynamics, BD particle modes, and induced operators require action and heavy-spectrum premises.	Algebraic adjoint proof plus action/heavy-mode certificates.
How is the μ-problem controlled?	The $\mathbb{Z}_4^R$ safety layer forbids perturbative $H_u H_d$ and allows nonperturbative generation.	Compactification realization.
How are moduli claims tested?	Dynamical stabilization requires stationarity and a physical Hessian bound. Target locking separately requires a precommitted augmented residual map and an exact or interval-certified isolation receipt. The proxy has rank zero on the 142-real-dimensional published pre-completion slice; no completed constraint map, stability certificate, or augmented rank is supplied.	Certified negative rank/isolation result plus a separate stability premise.
What is the string graviton?	A BRST-physical closed-string vertex with a positive-residue massless pole whose classical limit is the OPH transverse-traceless metric perturbation.	Quantum-particle receipt plus normalization match.
What fixes the critical dimension?	The critical-edge certificate must identify the sewn-edge carrier with the heterotic edge VOA; then heterotic criticality gives $D = 10$ and $c_{\mathrm{internal}} = 16$.	Edge transfer, OPE, grading, spin-structure, and no-coset receipts.

Theorem 22.1 (De Sitter/string compression theorem)**.** *On the OPH-correct branch, the de Sitter observer problem, the time-reversal clock-branch problem, the entropy-location problem, and the critical-string vacuum-selection problem reduce to one finite statement: find the observer-visible normal form of the OPH patch federation and then lift its sewn edge sector to a critical worldsheet presentation.*

Proof. The observer and clock problems are properties of record-bearing subfederations and their $\mathbb{Z}_2^T$ cycle data. The entropy problem is the edge-center capacity of the same finite static patch. The string problem is the critical completion of the sewn edge-sector partition. Complete response and endogenous transport fix the local Standard Model gauge Lie algebra. Conditional on the matrix current and declared matter category, the OPH normal form fixes the visible charge lattice and maximal faithful matter image used by all four questions. The apparently separate problems are projections of the same finite patch-federation normal-form problem. □

23 Conclusion

OPH gives a finite observer/record/overlap implementation layer for de Sitter holography. The same edge system admits a sewn-worldsheet language. Critical string theory enters as the effective completion of that language under the critical-edge premises. Conditional on those premises, the declared OPH selector target tests the Bouchard–Donagi one-Higgs heterotic Standard Model as a geometric zero-mode benchmark. Adding the MSSM $\mathbb{Z}_4^R$ safety layer defines the tested operator-safe proposal

$$BD^{\mathrm{OPH}}_{n=1,+} = BD^{\mathrm{OPH}}_{n=1} + \mathbb{Z}_4^R.$$

The sieve itself is fully specified through named premises, and the fixed target packet has no retunable target-side dial. The smooth published one-Higgs slice has 71 complex pre-completion moduli. The stated target-side proxy is constant in those directions, and any map defined on that full slice with the fixed five-real-coordinate comparison map has rank at most five. Completion constraints could reduce the physical dimension, but no such constraint locus or Jacobian is supplied. The published BD data contain neither a stabilized heavy spectrum nor a map from the supersymmetric compactification to the non-supersymmetric OPH low-energy branch. No threshold or spectrum conclusion follows. Therefore $BD^{\mathrm{OPH}}_{n=1,+}$ is not a selected string witness and functions only as a structural benchmark. The contract-conditional recovered OPH core has different premises; physical source binding, three-family attachment, and scalar multiplicity require additional premises.

This result does not prove that the BD compactification is physically inconsistent. It proves that the available data do not select it. No alternative string vacuum passes the certificate set in this paper. The strongest justified result is therefore an empty selected set for the stated catalogue, with BD as a structural candidate. Any positive result is local without the branch-cover theorem and catalogue-relative without candidate-universe coverage.

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